Dilutive Financing[*]

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Abstract

This paper presents a dynamic model of firm financing where firms use financial slack to reduce rent extraction by financiers with bargaining power. Financing is lumpy because it is optimal to bargain infrequently. Moreover, firms may finance 'early' before exhausting internal funds to bargain when their outside options are better. Financing rents are thus endogenous to firms' dynamic financing strategy. Firms with good financing alternatives raise financing early to reduce rents, whereas firms lacking such alternatives raise financing after exhausting funds to avoid frequently paying endogenously large rents. Investment irreversibility increases financing rents, and disproportionately so for less productive firms.

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1 Introduction

A pervasive feature of firm financing is that firms raise external financing infrequently in chunks rather than frequently in small increments. The canonical explanation in economics is based on the idea of fixed transaction costs, which dates back to Baumol (1952) and Tobin (1956).[1] In this paper, I present a new theory of dynamic firm financing that explains the infrequent "lumpy" financing as due to financier bargaining power. The theory also provides a natural explanation for another prevalent pattern which has received less attention in the literature: many firms raise external financing "early," that is, before running out of cheaper internal funds. These two key aspects of 'financial slack' jointly arise in my model as part of firms' optimal dynamic financing strategy to reduce rents that financiers are able to extract.

I consider an environment where a firm regularly needs financing, due to negative shocks or investment needs. While it may maintain internal funds in anticipation, these involve a carry cost.[2] The firm can raise funds from external financiers at any point, and there is no fixed transaction cost of financing. It is thus feasible, and first-best, to raise external financing incrementally whenever needed and keep zero internal funds.

The key element giving rise to the two dynamic patterns is a bargaining friction. A firm can bargain with financiers at any point to raise funds, but while it is bargaining with one financier, the firm cannot immediately locate *another* financier if the bargaining were to fail. This assumption of financiers' 'local monopoly' captures the idea that finding alternative sources of financing can prove challenging in practice because financiers are often rather specialized. For example, startups rely on venture capital funds with expertise for a specific industry and stage of the venture. Similarly, large firms resort to a handful of investment banks that address these firms' sizable funding needs with syndicated financing. In either case, it plausibly takes some time to switch to another financier in response to a failure of bargaining with the current financier.[3]

As a result, successful financing creates a positive and discrete surplus relative to a failure of bargaining, and financiers extract a portion of it as rents, thereby diluting firm value ex-ante. In response, firms choose to raise financing in excess of immediate funding needs, and maintain internal funds, in order to avoid needing to bargain with financiers and incur dilution too frequently. Thus, financing is optimally lumpy.

Moreover, firms may choose to raise financing early, before exhausting internal funds, in order to bargain when their outside options are better. With financial slack retained at bargaining, firms may avoid having their "back against the wall": even

[1] Baumol and Tobin (1989) note that Allais (1947) first proposed the same basic mechanism.

[2] 'Internal funds' may refer simply to cash reserves, but may also include, more broadly, any spare capacity for borrowing at a low interest rate, such as lines of credit.

[3] For modeling, one may consider search/informational frictions arising upon failed bargaining.

if financing were to fail, they could use these funds to cover losses, temporarily forestalling business failure while seeking alternative sources of financing. By raising funds before financing is really needed, firms can thus improve their outside options, thereby reducing the surplus from financing – and hence the rents that financiers can extract.

In sum, financial slack reduces dilution of firm value by improving firms' bargaining with financiers. Lumpy financing reduces the frequency of dilution, and early financing reduces its size. This, I argue, is what determines the dynamics of firm financing.

The point of departure is that financing costs – that is, financiers' rents – not only determine firms' optimal financing strategy, but also are determined by it. On one hand, the size of financing costs determines the optimal frequency of financing. On the other hand, the optimal *timing* of financing determines the size of financing costs. The first channel, when viewed in isolation, microfounds fixed transaction costs based on financier bargaining power. Both channels combined, the theory generates novel predictions on firm financing, as explained presently.

A key distinction of the model is that firms that raise financing early effectively offset financiers' bargaining power: by keeping sufficient financial slack when bargaining for funds, they reduce the surplus from which financiers extract rents to the point where equilibrium financing costs do not depend on how much bargaining power financiers possess. It may thus be incorrect to observe that some firms incur small financing costs and conclude that their financiers must have little bargaining power. To the contrary, the small costs may reflect that these firms optimally keep large financial slack when raising financing precisely because their financiers' bargaining power is substantial.

As a direct implication, the model predicts that firms with better access to financing may have more financial slack. Firms that are slow at accessing alternative sources of financing delay financing until running out of funds, because they cannot improve their outside options and reduce financing costs much by keeping financial slack when they bargain with financiers. In contrast, firms that can quickly access financing alternatives can greatly improve their outside options and reduce financing costs by raising financing when they have financial slack. In short, the incentive to raise financing early before exhausting financial slack *in*creases in the quality of firms' access to financing.[4]

Additionally, this paper suggests important cross-sectional implications when investment is irreversible, such as for intangible capital or during downturns.[5] Consider productive firms investing heavily. If bargaining for financing were to fail, they are able to forestall depletion of funds by reducing their investment expenses. Due to this ability to underinvest in response to a bargaining failure, these firms have good outside options vis-à-vis financiers when bargaining early with financial slack. In contrast,

[4]In canonical views, early financing is due to firms' exposure to fluctuations in financing conditions – and hence the incentive for it should decrease in the quality of access to financing.

[5]These conditions tend to coincide with difficulty of accessing alternative financiers.

unproductive firms that invest little are endogenously constrained in their ability to underinvest: if bargaining were to fail, they have little expenses to cut down on – and existing capital cannot be divested. Since these firms cannot improve their outside options as much, they optimally bargain after running out of funds. As such, investment irreversibility may disproportionately increase financing costs for less productive firms.

In terms of applicability, this theory naturally speaks to startup firms relying on venture capital funds, in the sense that entrepreneurs in practice do worry about bargaining and dilution that financing involves. Regarding large and established firms, it predicts that these firms have, in equilibrium, little reason to worry about bargaining with a financial institution such as an investment bank because they keep large financial slack. Across both cases, this paper makes a single claim: how much bargaining 'matters' in equilibrium depends endogenously on firms' optimal financial slack.

Methodologically, the framework I propose is tractable, and allows general characterization of optimal financing strategy. I characterize the unique Markov-perfect dynamic bargaining solution as simple (s, S) bounds, and establish comparative statics in bargaining power as well as access to alternative financing. As an aside, the model allows for a standard numerical solution given essentially as single-agent optimization, with an added binary state variable to track the firm's outside options.

The rest of the paper is organized as follows. Section 2 describes related literature. Section 3 presents the core mechanism with a stylized two-period model. Section 4 builds the main dynamic model in infinite time horizon, and Section 5 characterizes optimal financing strategy. Section 6 extends the framework with an investment choice and explores the heterogeneous effects of irreversibility. Section 7 concludes.

2 Related Literature

This paper makes contributions to four branches of the literature.

New framework of firm financial slack based on bargaining. As mentioned, the classic theory by Baumol (1952) and Tobin (1956) posits fixed transaction costs to explain households' cash-holdings. The theory has been applied to the firm context by Décamps, Mariotti, Rochet and Villeneuve (2011) and Bolton, Chen and Wang (2011) to rationalize lumpy firm financing. Bolton, Chen and Wang (2013) introduce fluctuations in fixed transaction costs to explain early financing with precautionary incentives.[6] In contrast, this paper suggests that both lumpy financing and early financing may arise from financier bargaining power, and generates a distinct prediction that firms with greater access to financing have greater incentives for early financing.

[6] The same mechanism is used by Alvarez and Lippi (2009) to study household money demand.

In Hugonnier, Malamud and Morellec (2014), firms face search frictions in financing. As such, when firms' matching rate to financiers increases, financing becomes, on average, both less lumpy and less 'early' – that is, less financial slack when financing is raised. In this paper, firms can always access financing but have potentially limited alternatives. When firms' access to financing alternatives improves, the model predicts that financing may simultaneously become less lumpy yet 'earlier.'

Empirically, this paper explains why large, established and creditworthy firms still maintain sizable financial slack, as reported by Graham (2022) and Albertus, Glover and Levine (2025). It is also consistent with Opler, Pinkowitz, Stulz and Williamson (1999), Bates, Kahle and Stulz (2009), Graham and Leary (2018) and Begenau and Palazzo (2021), who document that firms with steep growth have large financial slack.[7]

Endogenous timing of trades and endogenous dynamic outside options. The seminal work by Rubinstein and Wolinsky (1985) extends the foundational bargaining game of Rubinstein (1982) to endogenize agents' outside options through search frictions. In this class of models, the timing of trades is exogenous and driven by the random search process. Recently, McClellan (2024) endogenizes the timing of trades when the agent's outside option evolves exogenously. In contrast, both the timing of trades and the evolution of outside options are endogenous in my model.

This paper also relates to the literature on holdup such as Hart and Moore (1990) in that a first-best choice worsens agents' subsequent bargaining position. The contribution is that it studies optimal dynamic strategy given recurrence of holdup problems.

Kiyotaki, Moore and Zhang (2024) show that repeated rounds of bargaining dilute agents' payoffs and induce underinvestment. On the other hand, Lagos and Zhang (2022) study how outside options off-path may limit the counterparty's bargaining power on-path. These two mechanisms are parsimoniously unified in my model.

Complementing the literature on dynamic firm financing. The core premise of my model is that bargaining – that is, splitting of surplus – is recurrent, and the firm can choose when to bargain each time. This presumes a form of contract incompleteness in that the bargaining parties cannot commit to a long-term contract that would eliminate needs for any future bargaining. While this model restricts to equity trades for simplicity, such limited commitment is common in debt dilution literature such as Myers (1977), Brunnermeier and Oehmke (2013), Admati, DeMarzo, Hellwig and Pfleiderer (2018), DeMarzo and He (2021) and Donaldson, Koont, Piacentino and Vanasco (2024). This paper's contribution is twofold: it (i) emphasizes bargaining power – that is, how surplus is split – as another key determinant of financing dynamics, and (ii) offers a theory of equity dilution when the restriction to equity is taken more literally.

[7] See Supplemental Appendix C.

The paper builds on risk management literature such as Froot, Scharfstein and Stein (1993) by rationalizing financial slack in excess of investment needs. It also suggests that ease of liquidation may induce early refinancing of debt, instead of renegotiation upon default as in Bolton and Scharfstein (1996) and Hart and Moore (1998).

Capital liquidity, productivity and financing. The financial frictions proposed in this paper become severer when investment becomes more irreversible. This holds despite the absence of collateralized debt in the model, and arises because the ability to divest capital to obtain funds improves firms' outside options vis-à-vis financiers. Thus, this paper extends the results from the vast literature emphasizing how the resale price of capital affects borrowing constraints – including Kiyotaki and Moore (1997), Lorenzoni (2008), etc. – to general forms of firm financing including equity.

At the same time, when investment is highly irreversible, rent extraction by financiers may be endogenously lowered for productive firms that make large investment. This result contrasts with Rampini and Viswanathan (2010) where more productive firms with less capital are endogenously more constrained in financing their larger investment. Thus, this paper suggests that the cross-section of firm growth in intangible-intensive sectors with high irreversibility may be qualitatively different depending on whether financing mainly involves collateral constraints or bargaining frictions.

3 Core Mechanism

This section presents a stylized model that captures the core economic insights that due to financiers' bargaining power, firms may optimally raise financing infrequently and hence in a lumpy fashion, and also early – that is, before internal funds are exhausted.

Setup. There are two periods with three dates $t \in \{0, 1, 2\}$. There is no discounting. There is a project that lasts for the two periods. In each period, the project spends a unit of funds; with any less funds, it fails. The project yields a payoff $v > 2$ on terminal date $t = 2$. An agent called entrepreneur owns a firm that runs the project.[8]

The firm cannot self-finance the project. There are two financiers, one visiting the firm on date 0 and the other on date 1, who can provide it. Each of them has bargaining power and, if asked by the firm to provide funds, extracts a fraction $1 - \theta$, $\theta \in (0, 1)$, of the surplus from bargaining in addition to being compensated for the cost of funds normalized to unity. A financier cannot bargain with the firm or provide funds except on the date of his visit. Funds can be kept internally, but it requires a marginal carry

[8]The present setup gives a minimal environment that allows for multiple rounds of bargaining so that the core mechanism can be captured. The main model in Section 4 will entertain generic business profiles that allow for operating profits subject to occasional losses due to volatility.

cost $\varphi > 0$ to keep funds for one period.[9]

Denote h_t as the firm's holding of internal funds on date t; 'financial slack' is when h_t is positive. By backward induction, the firm raises from the second financier $\max\{0, 1 - h_1\}$ units of funds on date 1. If $h_1 \geq 1$, then there is no need to bargain with the second financier. If $h_1 < 1$, the project fails without more funds. The firm thus bargains with the second financier and pays him the cost of funds $1 - h_1$ plus his rent, which is a fraction $1 - \theta$ of the surplus from bargaining $v - (1 - h_1)$. Due to the carry cost $\varphi > 0$, $h_1 \in \{0, 1\}$ in equilibrium as long as $h_0 \leq 1$. Firm value on date 1 is, then, v if $h_1 = 1$, and $v - 1 - (1 - \theta)(v - 1) = \theta(v - 1)$ if $h_1 = 0$. Bargaining thus 'dilutes' firm value ex-ante: $\widehat{v} \equiv \theta(v - 1) < v - 1$ since $\theta < 1$.

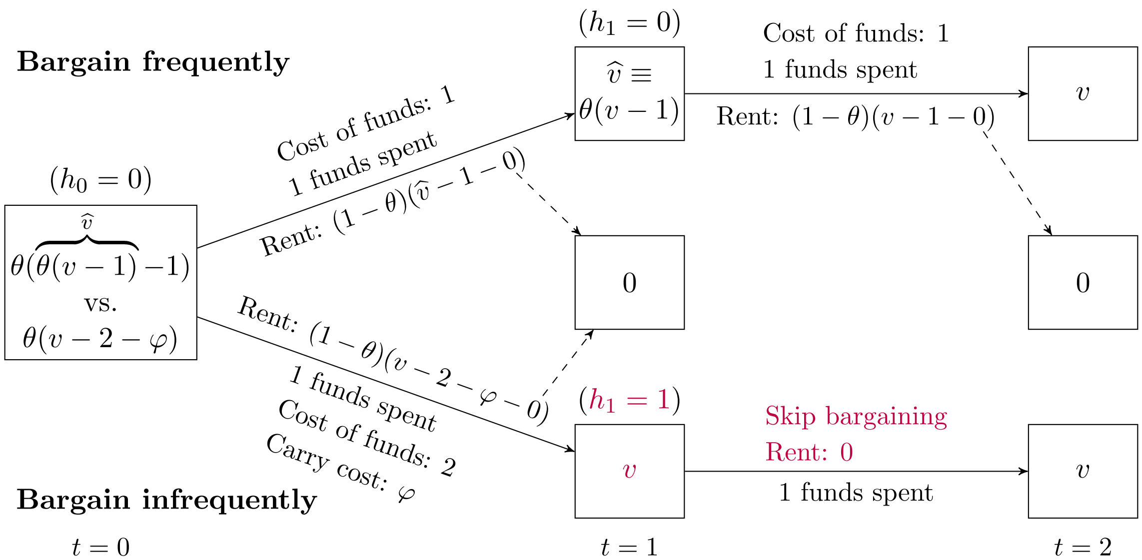


Figure 1: Frequency of bargaining

Rectangles show firm value at each node. The amount of internal funds on date t is h_t. Dashed arrows represent the firm's outside options – i.e., what happens if bargaining fails. Firm values and rents are determined as:

$$\text{Firm value}_t = \underbrace{\text{Firm value}_{t+1} - \text{Costs}_t}_{=\text{Continuation value}_t} - \text{Rent}_t, \quad \text{Rent}_t = (1-\theta)(\underbrace{\text{Continuation value}_t - \text{Outside option}_t}_{=\text{Bargaining surplus}_t}).$$

Frequency of bargaining and lumpy financing. Assume zero initial funds $h_0 = 0$. The entrepreneur chooses whether to bargain twice on both dates $t \in \{0, 1\}$ or once on date $t = 0$. If she bargains frequently (that is, twice), the entrepreneur raise a unit of funds from the first financier on date 0. No funds are kept internally $h_1 = 0$, so that firm value on date 1 is diluted $\widehat{v}$ due to the second bargaining. If she instead bargains infrequently (that is, once), she raises two units from the first financier. One is spent immediately, and the other is kept $h_1 = 1$ at the carry cost φ. Since there is no need

[9] This can represent either the entrepreneur's utility cost or decay by a factor of $\frac{\varphi}{1+\varphi} \in (0, 1)$.

to bargain with the second financier, firm value on date 1 is v.

Either way, the first financier, when bargaining with the firm on date 0, extracts as his rent a fraction $1-\theta$ of the difference between firm value on date 1 and the total costs on date 0, which include the cost of funds and, if applicable, carry cost.

The frequent bargaining maximizes net project value $v-2 > v-2-\varphi$, since the other choice involves a carry cost $\varphi > 0$. But the entrepreneur maximizes her own payoff, which is diluted firm value on date 0. Figure 1 describes the comparison. She optimally bargains infrequently if and only if $\theta\big(v-2-\varphi\big) \geq \theta\big(\theta(v-1)-1\big)$, or, equivalently,

$$(1-\theta)(v-1) \geq \varphi; \tag{1}$$

that is, if the rent that the second financier would extract $(1-\theta)(v-1)$ upon frequent bargaining exceeds the carry cost φ upon infrequent bargaining.[10]

Even though the carry cost φ makes the project less valuable, the entrepreneur may optimally bargain infrequently; by avoiding the need to bargain with the second financier, she ensures that her payoff from the project is diluted once rather than twice.

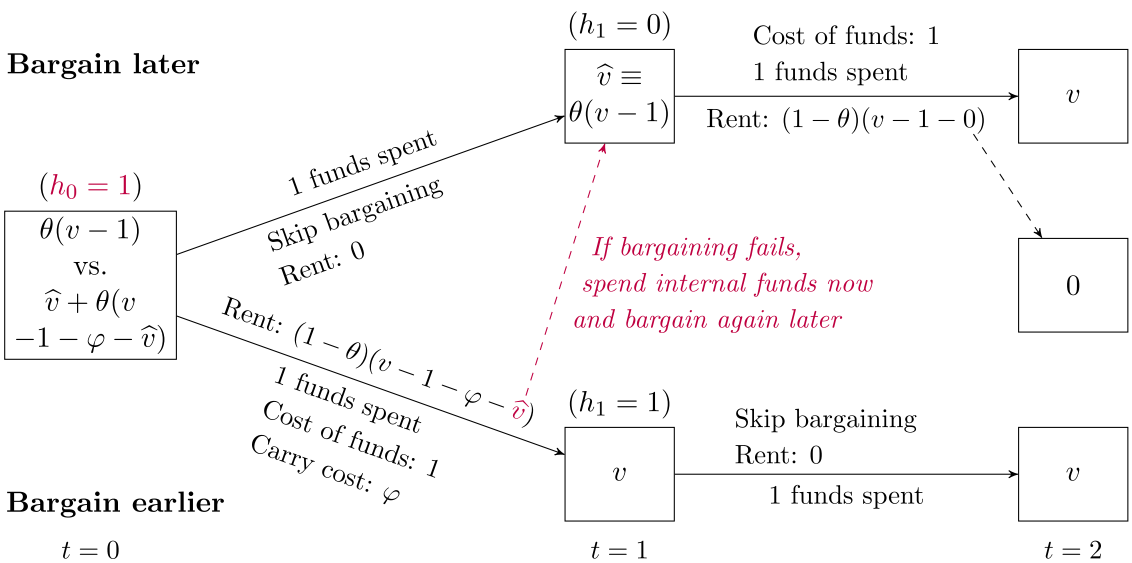


Figure 2: Timing of bargaining

Timing of bargaining and outside options. Next, suppose instead that the firm initially has one unit of funds $h_0 = 1$, so that it only needs to raise one unit for the second period. The entrepreneur needs to bargain once in any case, but must now choose when to bargain. If she bargains on date 1 with the second financier, then her payoff is $\widehat{v}$. Note that while bargaining with the second financier, the entrepreneur is at the brink of losing the project since there is no financial slack $h_1 = 0$. If instead she

[10]The entrepreneur may also optimally abandon the project if $\theta(v-1) \leq 1$ and $v-2-\varphi \leq 0$.

bargains earlier on date 0 with the first financier, she is no longer at the brink; even if bargaining were to fail on date 0, the entrepreneur can spend the internal funds $h_0 = 1$ now and bargain later with the second financier. Thus, the value of her outside option is $\widehat{v}$, instead of 0.

How does an improvement in the outside option affect the firm's value? The firm bargains with a financier to split the continuation value from bargaining, which is tomorrow's firm value minus today's cost of funds and carry. But the firm has already secured the outside option value even if the bargaining were to fail. It is thus only the remainder – 'bargaining surplus' – that actually gets split according to the bargaining weights $(\theta, 1-\theta)$ with the financier. That is,

$$\begin{aligned}\text{Firm value}_t &= \text{Continuation value}_t - \text{Rent}_t \\ &= \text{Continuation value}_t - (1-\theta)\big(\underbrace{\text{Continuation value}_t - \text{Outside option}_t}_{=\text{Bargaining surplus}_t}\big) \\ &= \text{Outside option}_t + \theta\big(\text{Continuation value}_t - \text{Outside option}_t\big).\end{aligned}$$

Therefore, an increase in the firm's outside option when it bargains earlier, if continuation value is fixed, raises firm value by a fraction $1-\theta$. Of course, earlier bargaining reduces continuation value due to carry cost φ, reducing firm value by a fraction θ.

As Figure 2 outlines, the entrepreneur optimally bargains earlier if and only if $\widehat{v} + \theta(v - 1 - \varphi - \widehat{v}) \geq \theta(v-1)$, or, equivalently,

$$(1-\theta)\widehat{v} \geq \theta\varphi; \tag{2}$$

i.e., gain from a better outside option $(1-\theta)\widehat{v}$ outweighs her loss from carry cost $\theta\varphi$.

Discussion. Financial slack may endogenously arise on date 1, despite the carry cost, if it is optimal to bargain infrequently rather than frequently or earlier rather than later. The entrepreneur bargains infrequently by raising enough at a time to sustain multiple periods without needing to bargain and pay rents, thereby reducing the frequency of dilution. She may also optimally raise funds earlier to bargain when her outside options are better, thereby reducing the size of dilution.

These results arise because of bargaining frictions. While the firm cannot immediately raise funds from another financier if financing with one financier were to fail, nothing in the model would make the firm actually undergo this phase in equilibrium, because it is suboptimal for either party to let financing fail.[11] But the negative consequences that would arise if financing *were to* fail determine the outside options at bargaining, and hence the surplus from successful financing. Because financiers have

[11] Supplemental Appendix SA.1 discusses setups where multiple financiers visit each period.

bargaining power $1 - \theta > 0$, the firm does not fully internalize this surplus. Therefore, its optimal choice of when to raise funds may fail to be first-best.

Additionally, the second result of early bargaining leads to a distinct comparative statics. Again given the initial endowment $h_0 = 1$, now assume that if bargaining were to fail on date 0, the second financier cancels his visit on date 1.[12] Essentially, the firm's access to financing is now 'weaker' because it gets lost for good upon a bargaining failure. Then, the firm has its "back against the wall" vis-à-vis either financier, even if it bargains early with financial slack: the project fails if bargaining fails because the firm can no longer raise additional funds. Since the firm's outside option value is now invariably zero on both dates $t \in \{0, 1\}$, the left-hand side of Inequality (2) is zero, such that early bargaining is suboptimal for any parameters. In short, it is only firms with *better* access to financing that may optimally bargain early with financial slack.

The above analysis captures the core of the mechanism that I propose, but it is incomplete. The initial funds $h_0 \in \{0, 1\}$ should reflect the firm's preceding choice, and so must be endogenized. In particular, although I separately discussed infrequent bargaining and early bargaining for clarity of each insight, they should interact in principle.[13] Therefore, I transition to a fully dynamic setup in infinite horizon for more complete analysis.

4 Model

This section sets up the main model of firm financing subject to bargaining frictions.

Environment. Time is continuous and infinite $t \in [0, \infty)$. All agents are risk neutral and have a common time discount rate $\rho > 0$. Agents called insiders own a firm to run a business. The business has an exogenous cash flow profile, given in Section 4.1. The firm may hold internal funds (or 'funds') $h_t \geq 0$, to which cash flow accrues. Internal funds yield a return $r < \rho$.[14] The spread $\varphi \equiv \rho - r > 0$ is the associated carry cost.

Insiders may frictionlessly receive a nonnegative dividend. If internal funds are depleted without immediate financing, the business fails with zero salvage value.

Financing bargaining. To avoid business failure, the firm needs regular financing. Insiders are assumed to be penniless and hence must raise funds for the firm from homogeneous agents called financiers. Insiders can choose when to bargain à la Nash with financiers for funds.[15] Bargaining weights for insiders and financiers are $\theta \in (0, 1)$

[12] One way to microfound this modification is that failed bargaining is seen as a negative signal.

[13] If early bargaining at $t = 0$ is not optimal, the firm at $t = -1$ with no funds $h_{-1} = 0$ may find it optimal to raise three units instead of two from the financier on that date.

[14] Internal funds can, via a simple reformulation of cash flow in Section 4.1, represent slack against borrowing constraint in a risk-free instantaneous lending market with interest rate $r < \rho$.

[15] Other axiomatic bargaining solutions, such as Kalai and Smorodinsky (1975) and Kalai

and $1-\theta$, respectively. In return for providing funds, financiers receive a proportional ownership stake in the firm.

To focus on the bargaining-driven dynamics of financing, I do not separate the timing of bargaining from the timing of financing: at any $t \geq 0$, no agents can commit to a future transaction at $t' > t$. For further simplicity, I assume that financiers are deep-pocketed but upon financing, they become penniless insiders.

Outside options. If insiders can walk away from bargaining and immediately find alternative financiers, bargaining is trivial due to perfect competition among financiers. I therefore assume financier local monopoly: if bargaining with one group of financiers were to fail, it takes time for insiders to find another group of financiers to bargain with and raise funds from. I call this time lag 'exclusion,' which can be permanent or temporary. If permanent, excluded firm insiders continue the business until funds are depleted, at which point the business fails. Generally, excluded insiders are 're-included' at a Poisson arrival rate $\gamma \geq 0$ that parametrizes the accessibility of alternative financing – specifically, the speed of accessing such alternative sources. Thus, insiders face a stochastic time lag in finding another financier upon a failed bargaining. Re-inclusion means restoring the 'access,' i.e., the ability to freely choose when to raise funds from financiers.

Comments on interpretations. First, the overall setup can be viewed as a search friction but only off the equilibrium path. This is, arguably, rather plausible. Corporate financial managers can forecast whether funds will be needed soon and start engaging with financial investors in advance. If, however, a disagreement during the bargaining were to derail the planned financing, they would need to start looking for other options for financing.[16] Even if no disagreement arises, what might happen if one did determines what does happen in terms of bargaining outcomes.

Second, the assumption that deep-pocketed financiers become penniless upon financing can be viewed as representing in a simplified way that each financier has a limited funding capacity but can form a syndicate if needed such that optimal amount of financing can always be secured on demand. One may also consider, without loss, insiders bargaining with an intermediary that syndicates perfectly competitive financiers.

Lastly, the restriction to one-off trades rather than long-term state-contingent contracts can be microfounded through either non-verifiability of firms' internal funds h_t or inability to prevent renegotiation ex post given each financier's bargaining power.

(1977), are equivalent to Nash (1950) (under suitable generalizations) in this setup due to the assumptions of deep-pocketed financiers and risk-neutral preferences.

[16] One may regard the present assumption as a stylized version of search frictions where a match is both durable until actual financing and exclusive. Informational frictions upon failed financing, as discussed at the end of Section 3, may also be considered.

4.1 Cash flow

The business has an exogenous cash flow profile. Consider two concrete examples:

1. The business is a 'startup' that incurs a fixed flow expense $\kappa\ dt$ with $\kappa > 0$ ('burn rate'), until succeeding upon a stopping time τ at a Poisson rate $\lambda > 0$. Upon success, the business earns a terminal payoff $\Pi > \frac{\kappa}{\lambda}$ and ends.
2. The business is an 'operating firm' that indefinitely earns an average flow profit $\pi\ dt$, where $\pi > 0$, but subject to occasional losses due to volatility $\sigma\ dB_t$, where $\sigma > 0$ and B_t is a standard Brownian motion.

Generically, time-t cash inflow is $\mu\ dt + \sigma\ dB_t$, where $\mu \in \mathbb{R},\ \sigma \geq 0$; for startups, $\mu \equiv -\kappa < 0$ and for operating firms, $\mu \equiv \pi > 0$. The business may 'succeed' upon a stopping time τ at a Poisson rate $\lambda \geq 0$ with a terminal dividend $\Pi + h_\tau$, $\Pi \in \mathbb{R}$.

Assumption 1. *$\mu + \lambda\Pi > 0$. If $\sigma = 0$, then $\mu < 0$ and either $r \leq 0$ or $\frac{\mu+\lambda\Pi}{\rho+\lambda} < \frac{|\mu|}{r}$.*

The first part gives a positive net present value of the business $\frac{\mu+\lambda\Pi}{\rho+\lambda} > 0$ in the absence of frictions. The second part ensures that external financing is regularly needed.[17]

4.2 Value functions and bargaining for financing

I consider Markov-perfect equilibrium. By linear preference, optimal dividend policy is $\max\{0, h_t - \overline{h}\}$ above a threshold $\overline{h} \geq 0$. Letting $V(h)$ denote equilibrium value of insiders (including ones that previously were financiers) given funds h, $\overline{h}$ solves dividend optimality $V'(\overline{h}) = 1$, i.e., smooth pasting, and super contact $\sigma > 0 \implies V''(\overline{h}) = 0$.

Suppose that at a given point, firm insiders have chosen to bargain with financiers for funds. Let $V_o(h)$ denote insiders' reservation value ('outside option if bargaining fails') given funds h, and $x \in [0, 1]$ their retained ownership. Nash bargaining solves

$$\max_{x \in [0,1],\ \overline{h} \geq 0} \Big(xV(\overline{h}) - V_o(h)\Big)^{\theta} \Big((1-x)V(\overline{h}) - (\overline{h} - h)\Big)^{1-\theta}.$$

The solution is straightforward: $\overline{h} \in \arg\max_{\widetilde{h}}\ V(\widetilde{h}) - \widetilde{h}$, and

$$x(h)V(\overline{h}) = V_o(h) + \theta\Big(V(\overline{h}) - (\overline{h} - h) - V_o(h)\Big). \tag{3}$$

The funding target $\overline{h}$ maximizes net firm value $V(\overline{h}) - \overline{h}$.[18] Ownership is split in a way that insiders' retained value from bargaining $x(h)V(\overline{h})$ given h equals their reservation

[17] Even without volatility $\sigma = 0$, loss occurs with positive probability $\mu < 0$ (i.e., certainty), and fully covering losses with returns on internal funds $h_t \geq |\mu|/r$ is suboptimal.

[18] The funding target, taken as the infimum of the arg max, coincides with the dividend payout threshold because both equalize marginal value of funds with marginal cost of financing. This identity may fail given transaction costs of dividend payout, but this paper's results are robust.

value $V_o(h)$ plus a fraction θ of the bargaining surplus $V(\overline{h}) - (\overline{h} - h) - V_o(h)$; the other fraction $1 - \theta$ of it goes to financiers as their rents.

Algebraically, financing is optimal at h if and only if firm value is given by $V(h) = x(h)V(\overline{h})$. The presence of $\theta < 1$ in $x(h)V(\overline{h})$ shows that the firm's value function V is diluted in anticipation of financiers' rent extraction from optimally-timed financing.

4.3 Dynamic programming

The simplifications in the setup make the financing history payoff-irrelevant, allowing for a straightforward dynamic programming for the pair of value functions (V, V_o) that depend only on the amount of internal funds h.

Boundary conditions. First, dividend optimality gives $V'(\overline{h}) = 1$ by smooth pasting and, if $\sigma > 0$, $V''(\overline{h}) = 0$ by super contact.[19] Second, the business fails if funds are depleted without access to financing, that is, $V_o(0) = 0$. Third, insiders have access to financing, that is, $V(h) \geq V_o(h) + \theta\big(V(\overline{h}) - (\overline{h} - h) - V_o(h)\big)$, where equality means optimal financing; since financing must be optimal at $h = 0$, $V(0) = \theta\big(V(\overline{h}) - \overline{h}\big)$.

HJB equations. When insiders are neither receiving dividends nor raising financing, their equilibrium value function V and reservation value function V_o solve

$$\begin{aligned} \rho V(h) - rhV'(h) &= \mu V'(h) + \frac{1}{2}\sigma^2 V''(h) + \lambda\big(\Pi + h - V(h)\big), \\ \rho V_o(h) - rhV_o'(h) &= \mu V_o'(h) + \frac{1}{2}\sigma^2 V_o''(h) + \lambda\big(\Pi + h - V_o(h)\big) + \gamma\big(V(h) - V_o(h)\big). \end{aligned} \tag{4}$$

The last term in Equation (4) reflects that if financing were to fail such that insiders' value function becomes V_o, they can regain access to financing – that is, revert to V – at a Poisson rate γ. To emphasize, financing does not fail in equilibrium.

5 Analysis

This section presents the analysis of the equilibrium – which exists and is unique.

Lemma 1 (Existence and uniqueness)**.** *There exist unique V and V_o, the equilibrium value function of insiders and their reservation value function.*[20]

[19]The same holds for V_o with respect to the corresponding dividend threshold $\overline{h}_o$.

[20]The proof in Appendix A.1 leverages the two-step recursive structure shown in Section 4.3: (Markov-perfect) equilibrium value function V depends on itself because (i) V depends on V_o through bargaining θ, and (ii) V_o depends on V through re-inclusion γ. The expected discounting at rate $\rho > 0$ due to the time delay in re-inclusion $\gamma < \infty$ makes this recursion strictly contractionary. Accordingly, V exists as a unique fixed point by contraction mapping theorem. The second step of the recursion then determines V_o. The proof is graphically illustrated in Supplemental Appendix SA.2.

Given the assurance, Sections 5.1 and 5.2 discuss lumpy financing and early financing. Section 5.3 illustrates how financial slack reduces dilution, and Section 5.4 characterizes equilibrium financing costs. Section 5.5 presents comparative statics.

5.1 Lumpy financing

Let $\overline{h} = \inf\{\arg\max_h \; V(h) - h\}$.[21] Define $B \subset [0, \overline{h}]$ as the set of internal funds h at which it is optimal in equilibrium to bargain with financiers for funds: $h \in B$ if and only if

$$V(h) = x(h)V(\overline{h}) = V_o(h) + \theta\big(V(\overline{h}) - (\overline{h} - h) - V_o(h)\big); \tag{5}$$

obviously, $0 \in B$. Equation (5) gives an expression for financiers' rents as

$$(1-\theta)\Bigg[\underbrace{\Big(V(\overline{h}) - (\overline{h} - h) - V(h)\Big)}_{\equiv J(h)} + \underbrace{\Big(V(h) - V_o(h)\Big)}_{\equiv E(h)}\Bigg]. \tag{6}$$

Above, $J(h)$ is the joint surplus from financing when insiders choose to enter into bargaining with financiers, and $E(h)$ is insiders' loss due to exclusion: if bargaining fails once they have entered into it, insiders get $V_o(h)$ instead of $V(h)$. Financiers receive as their rents a fraction $1-\theta > 0$ of not just $J(h)$ but also $E(h)$. Since $\gamma < \infty$, exclusion always involves a discrete loss, that is, $E > 0$ globally (which will be shown shortly). Therefore, insiders cannot eliminate financing rents even in the limit by raising financing at h arbitrarily close to $\overline{h}$. At the same time, $h \to \overline{h}^-$ implies infinite financing frequency, that is, paying $(1-\theta)(J(h) + E(h)) \to (1-\theta)E(\overline{h}) > 0$ infinitely often. Exact incremental financing, therefore, is not optimal in continuous time limit.

Note that financiers cannot extract rents if either $\theta = 1$ or $\gamma \to \infty$. It is obvious from Equation (6) that $\theta = 1$ eliminates financiers' rents. Fix $\theta \in (0,1)$ but let $\gamma \to \infty$ so that $E(h) \to 0$ pointwise for $h > 0$. Note that for any $\gamma < \infty$, $h \to \overline{h}$ gives $J(h) \to J(\overline{h}) = 0$.[22] Therefore, when $\gamma \to \infty$, financiers' rents $(1-\theta)\big(J(h) + E(h)\big)$ vanish as financing occurs infinitely close to the funding target $\overline{h}$. As expected (and demonstrated shortly in Section 5.5), there is no financial slack in the absence of rent extraction by financiers, that is, either $\theta = 1$ or $\gamma \to \infty$.

To summarize, the present theory predicts 'lumpy' and infrequent financing if and only if bargaining is nontrivial. This feature is demonstrated by the first main result below and its proof.

Proposition 1. *Financing is lumpy and intermittent, i.e.,* $\sup B < \overline{h}$.

[21] When $h_t > \overline{h}$, the firm immediately pays out dividends $h_t - \overline{h}$ to insiders. Thus, $h \mapsto V(h) - h$ is constant above $\overline{h}$, and therefore choosing the funding target $\overline{h}$ as the infimum is without loss.

[22] It is easily shown that as $\gamma \to \infty$, $J(h) \to 0$ for $h > 0$ conditional on financing being optimal.

Proof. Suppose not, that is, for any sufficiently small $\varepsilon > 0$, financing at $h = \overline{h} - \varepsilon$ in the amount ε is optimal. By Equation (5),

$$V(\overline{h} - \varepsilon) = x(\overline{h} - \varepsilon)V(\overline{h}) = \theta\big(V(\overline{h}) - \varepsilon\big) + (1 - \theta)V_o(\overline{h} - \varepsilon).$$

Letting $\varepsilon \to 0^+$ gives $V(\overline{h}) = \theta V(\overline{h}) + (1-\theta)V_o(\overline{h})$ by continuity (see Appendix A.1). Since $\theta < 1$, this is equivalent to $V(\overline{h}) = V_o(\overline{h})$. But by Assumption 1, there is a finite time interval over which, without financing, internal funds $\overline{h}$ get depleted with nonzero probability. Also, over any finite time interval, re-inclusion fails to occur with nonzero probability since $\gamma < \infty$. In sum, there is nonzero probability that today's exclusion causes business failure within a finite time interval. Without exclusion, positive surplus is retained at depletion since $\theta > 0$. Hence, $V(\overline{h}) > V_o(\overline{h})$, a contradiction. $\square$

5.2 Early financing

Next, let us analyze insiders' financing strategy $B \ni 0$. The following lemma establishes that it is an interval from zero: that is, optimal financing strategy is monotone in internal funds h. While this is a technical result, I include a partial proof in the article because its derivation points at the core of the economic mechanism.

Lemma 2 (Monotone financing strategy)**.** *If $h \in B$, then $[0, h] \subset B$.*

Proof. Suppose $B \setminus \{0\}$ is nonempty. Define F as the payoff function when insiders immediately raise financing given any funds h and subsequently follow optimal strategy:

$$F(h) \equiv x(h)V(\overline{h}) = \theta\big(V(\overline{h}) - (\overline{h} - h)\big) + (1-\theta)V_o(h) \leq V(h). \tag{7}$$

Immediate financing is optimal on the set B, and hence is preferred to an instantaneously delayed financing: for $h \in B \setminus \{0\}$,

$$\rho F(h) - rhF'(h) \geq \mathcal{H}(F)(h) \equiv \mu F'(h) + \frac{1}{2}\sigma^2 F''(h) + \lambda\big(\Pi + h - F(h)\big). \tag{8}$$

Also, dividend optimality conditions for V imply that

$$\rho V(\overline{h}) - r\overline{h} = \mathcal{H}(V)(\overline{h}) = \mu + \lambda\big(\Pi + \overline{h} - V(\overline{h})\big). \tag{9}$$

Equation (7) implies that $F'(h) = \theta + (1-\theta)V_o'(h)$ and $F''(h) = (1-\theta)V_o''(h)$. Thus,

$$\mathcal{H}(F)(h) = \theta\mathcal{H}(V)(\overline{h}) + (1-\theta)\mathcal{H}(V_o)(h). \tag{10}$$

Substituting (4), (7), (9), (10) into (8) cancels out $\mathcal{H}(F)(h)$ and gives: for $h \in B \setminus \{0\}$,

$$(1-\theta)\gamma\big(V(h) - V_o(h)\big) \geq \theta\varphi\big(\overline{h} - h\big), \tag{11}$$

which is equivalent to Inequality (8). For the rest of the proof, see Appendix A.2. □

Inequality (11) captures the core trade-off associated with early financing. Suppose that financing is optimal at $h_t = h > 0$. This implies that insiders prefer bargaining earlier now to delaying bargaining by an instant dt as a one-shot deviation.[23] Two factors are relevant in this comparison. On one hand, there is a carry cost when funds $\overline{h} - h_t$ come in at t instead of at $t + dt$. This does not affect insiders' payoff from delaying financing until $t + dt$, but reduces continuation value from time-t bargaining $V(\overline{h}) - (\overline{h} - h_t)$ by $\varphi(\overline{h} - h_t)\ dt$ in anticipation. When insiders bargain earlier at t, they incur a fraction θ of this loss since financiers bear the rest $1 - \theta$ as a reduction in rents. This is the right-hand side. On the other hand, there is a chance of finding alternative financiers during $(t, t + dt]$. This, again, does not affect the payoff from delaying financing until $t + dt$, but raises insiders' time-t reservation value $V_o(h_t)$ by $\gamma(V(h_t) - V_o(h_t))\ dt$ in anticipation. When insiders bargain earlier at t, they gain by a fraction $1 - \theta$ from this improvement in outside options since financiers' rents are reduced by as much. This is the left-hand side.[24]

Lemma 2 allows the equilibrium to be fully characterized by '(s, S)' bounds $(\underline{h}, \overline{h})$, where $\underline{h} \equiv \sup B$. When $\lim_{s \to t^-} h_s = \underline{h}$, insiders raise financing in the amount $\Delta h \equiv \overline{h} - \underline{h} > 0$ so that $h_t = \overline{h}$. Henceforth, I refer to $\underline{h}$ both as 'financing threshold' – to address the dynamics of financing – and 'funding reserve' – to focus on the amount of financial slack when financing is raised. I say that insiders engage in 'early financing' if $\underline{h} > 0$, that is, if insiders raise financing before running out of funds. I also refer to $\overline{h}$ and Δh as 'funding target' and 'financing amount,' respectively.

Proposition 2 (Monotonicity of early financing)**.** *Given the other parameters, there exists* $\underline{\gamma} \in (0, \infty)$ *such that early financing is optimal, i.e.,* $\underline{h} > 0$*, if and only if* $\gamma > \underline{\gamma}$*. Early financing is thus never optimal, i.e.,* $\underline{h} = 0$*, if* $\gamma = 0$*.*

Proposition 2 formalizes the logical implication from the discussion of Lemma 2 that the only reason that insiders would choose to finance early, $\underline{h} > 0$, is that if financing were to fail, off the equilibrium path, insiders could use this funding reserve to cover losses while seeking alternative sources of financing to prevent business failure. Importantly, it is not that insiders worry about bargaining failure per se – bargaining

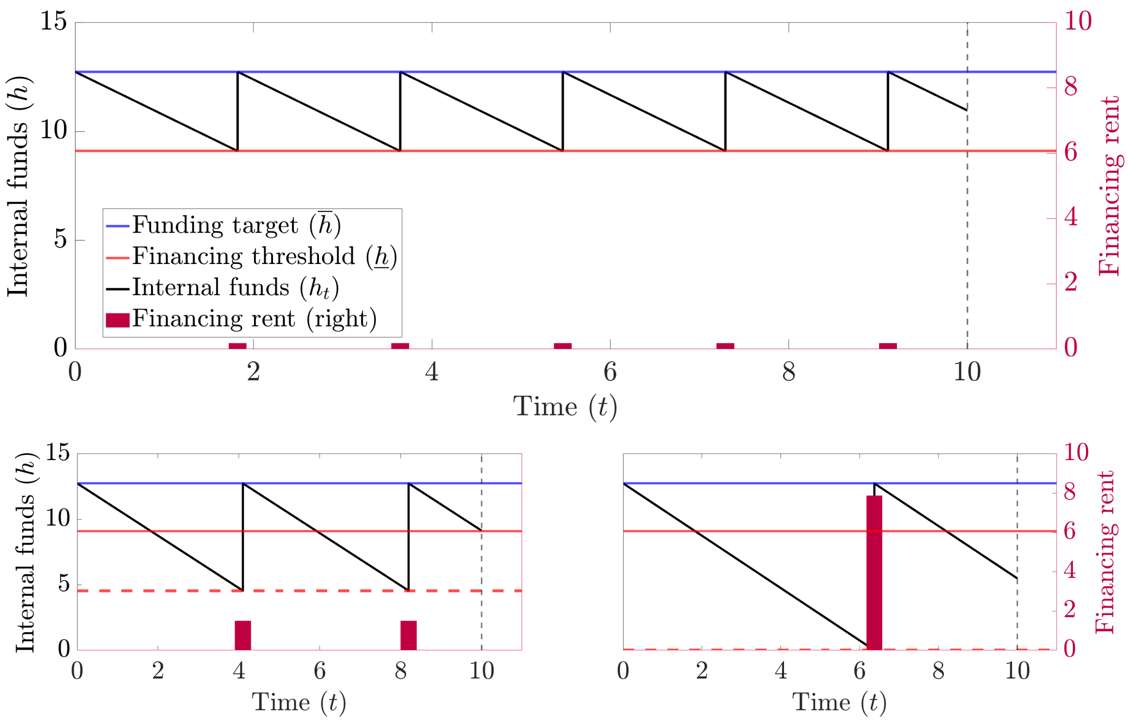
[23]This comparison is marginal in time (t versus $t + dt$), rather than in funds (h versus $h + dh$).

[24]Inequality (11) coincides with Inequality (2) in the stylized model of Section 3, with one innocuous generalization: if bargaining fails on $t = 0$, the second financier cancels his visit with probability $1 - \gamma$, $\gamma \in [0, 1]$. Earlier bargaining raises the firm's reservation value by $\gamma\big(\widehat{v} - 0\big)$.

never fails in the model – but rather that a better preparedness against it means better outside options when they bargain with financiers. Thus, the quicker insiders can find alternative financiers upon a failure of financing, the more they can reduce financing rents by raising financing early. Thus, early financing, as a binary choice $\mathbb{1}(\underline{h} > 0)$, is weakly *in*creasing in firms' access to alternative financing γ.

5.3 Illustration

I now graphically explain how financial slack reduces dilution. Consider a startup that incurs a fixed expense $\kappa\ dt$ until success at Poisson rate λ with a terminal payoff Π. Baseline parameters are: $(\rho, r) = (0.05, 0)$, $(\theta, \gamma) = (0.5, 1)$ and $(\kappa, \lambda, \Pi) = (2, 0.1, 50)$.



(a) Halved funding reserve $\widetilde{h} = 0.5\underline{h}$ (b) Finance only at depletion $\widetilde{h} = 0$

Figure 3: Financial slack and dilution

The plots describe a startup with parameters $(\rho, r) = (0.05, 0)$, $(\theta, \gamma) = (0.5, 1)$, $(\kappa, \lambda, \Pi) = (2, 0.1, 50)$. The graphs in solid black tracks the history of internal funds h_t until business success – at $t = 10$ in this sample. Financing (i.e., h_t jumping from financing threshold $\underline{h}$ to funding target $\overline{h}$) involves dilution given in magenta bar. The bottom subplots depict suboptimally lower financing thresholds. An operating firm $\pi\ dt + \sigma\ dB_t$ with $(\pi, \sigma) = (1, 2)$ yields the same patterns, including how counterfactual financing thresholds affect the size of dilution.

Size-frequency tradeoff. Figure 3 illustrates the relationship between financial slack and dilution. In the main plot that shows optimal financing strategy, insiders raise financing once every $\frac{\Delta h}{\kappa} \approx 1.8$ periods (until success) and 'very early' with a large funding reserve $\underline{h} \approx 9.1$. At each financing, insiders incur small dilution: 0.18 in value.

If this cost were 'fixed,' the strategy that the main plot illustrates would be strictly dominated by further delaying financing with lower thresholds $\widetilde{h} < \underline{h}$, as the cost would be less frequently incurred. The bottom two subplots, Figures 3a and 3b, show that these deviations are indeed not optimal.[25] Although frequency decreases, the size of dilution endogenously magnifies, to 1.5 with $\widetilde{h} = 0.5\underline{h}$ and even up to 7.88 with $\widetilde{h} = 0$.

Why is dilution magnified when insiders raise financing with smaller financial slack? The answer lies in insiders' outside options. In Figure 4, I decompose valuation on the vertical axis, assuming a one-shot strategy of immediate financing at each h across $[0, \overline{h}]$ on the horizontal axis.[26] The rightmost edge in blue is the funding target $\overline{h}$, and the red vertical line the financing threshold $\underline{h}$.

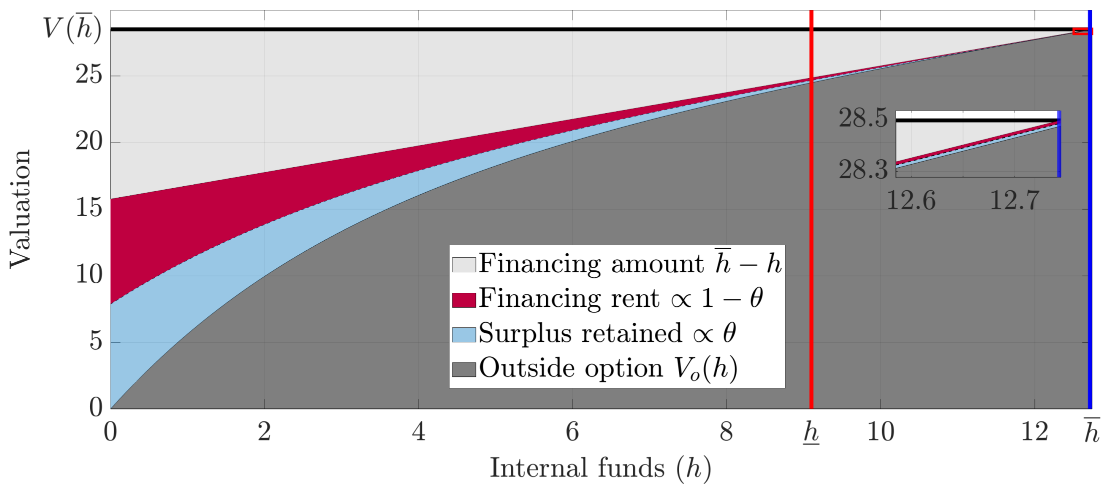


Figure 4: Financial slack, outside options, and the size of dilution

This plot addresses the same startup example as in Figure 3. It shows value decomposition when insiders with funds $h \in [0, \overline{h}]$ choose to raise financing – suboptimally on $(\underline{h}, \overline{h}]$. Continuation value from bargaining is $V(\overline{h}) - (\overline{h} - h)$, which is the height of the bottom of the light gray area, and outside option value is the height of the dark gray area. The surplus from financing is in between the two, which is split, by ratio $(1 - \theta : \theta)$, to financiers' rents in magenta and insiders' retained surplus in light blue.

The solid black line at the top of Figure 4 is firm value upon financing $V(\overline{h})$. To attain this 'post-money' value, financiers must provide financing $\overline{h} - h$, represented as the height of the light gray area right below it; this amount decreases in h at a unit slope. Financing surplus, however, is not simply the difference between post-money value $V(\overline{h})$ and 'money' $\overline{h} - h$. Insiders possess outside options $V_o(h)$ at bargaining, represented as the height of the dark gray area. Even if financing were to fail, insiders with financial slack may still seek alternative funding sources at Poisson rate $\gamma = 1$

[25]For ease of graphical illustration, I assume that these deviations – although appearing as recurrent in the two subplots – are taken by the agents as 'one-shot' until each time of financing. Anticipated deviations post financing would affect firm value ex-post, endogenously altering the 'schedule' of financing rents ex-ante as shown in Figure 4.

[26]This is what F in the proof of Lemma 2 represents, and is suboptimal on $(\underline{h}, \overline{h}]$. One distinction is that the present comparison is in terms of funds h, rather than time t.

to prevent business failure. But if financing were to fail without slack, the business promptly fails. Thus, the value of outside options V_o rises steeply in financial slack h.

Therefore, the financing surplus, which is the two colored areas in the middle, decreases in financial slack. This is what gets divided, according to the ratio $(1-\theta : \theta)$, into financiers' rents in magenta and insiders' surplus retention in light blue.

In short, financial slack monotonically reduces the size of dilution. But then, why is the optimal funding reserve $\underline{h}$ not even higher? The answer is the frequency. Given funding target $\overline{h}$, a higher financing threshold $\underline{h}$ decreases financing amount Δh and thus increases the frequency of dilution, while its size no longer decreases as steeply.[27]

Access to alternative financing γ. When is early financing not optimal? As Proposition 2 shows, it is not optimal when insiders cannot quickly access alternative financing $\gamma \leq \underline{\gamma}$. Then, financial slack does not improve insiders' outside options enough to justify the increases in carry cost and financing frequency.

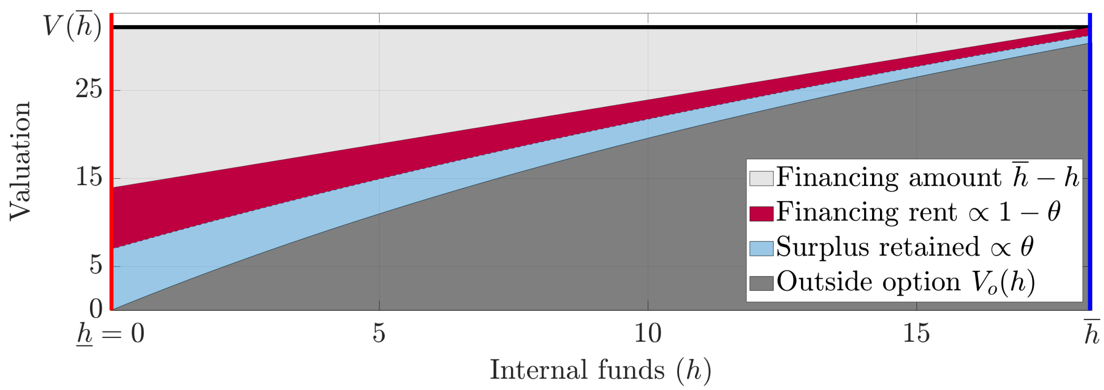


Figure 5: Poor access to alternative financing $\gamma = 0$

The height of the plot is adjusted so that slopes are visually comparable to those in Figure 4.

Figure 5 illustrates it with $\gamma = 0$ for an otherwise identical startup. Financial slack increases insiders' outside options at a much lower slope, since insiders cannot access alternative financing if financing were to fail. Consequently, financial slack does not reduce the size of dilution enough to compensate for its increased frequency. These insiders instead delay financing until funds are depleted and, when they do raise funds, raise a much larger amount $\Delta h \approx 18.27 > 3.64$ to reduce the frequency of dilution.

5.4 Endogenous financing costs

As illustrated, there are two types of optimal financing strategies, depending on whether financing is early. Since financing costs are endogenous to financing strategy, financing

[27]There is a gap even at the top $V(\overline{h}) - V_o(\overline{h}) > 0$, which is a mere 0.022 but certainly nonzero. As Proposition 1 shows, this non-vanishing loss from exclusion induces lumpy financing.

costs in equilibrium look qualitatively different in each case, as formalized below.

Lemma 3 (Endogenous financing costs)**.** *In equilibrium, financiers extract rents from insiders through financing in the amount of:* $(1-\theta)\big(V(\overline{h})-\overline{h}\big)$ *if insiders delay financing until funds are depleted, i.e.,* $\underline{h}=0$*; and, if insiders finance early, i.e.,* $\underline{h}>0$*,*

$$\frac{\varphi}{\gamma}\Delta h+\frac{1}{2}\frac{\sigma^2}{\theta\gamma}\Big(V''^{(+)}(\underline{h})-V''^{(-)}(\underline{h})\Big)\geq\frac{\varphi}{\gamma}\Delta h. \tag{12}$$

Proof. Financiers' rent when $\underline{h}=0$ is from (3). For (12), rearrange $\rho V(\underline{h})-r\underline{h}V'(\underline{h})=\mathcal{H}(V)(\underline{h})$ (using the right second derivative when $\sigma>0$) following the steps in the proof of Lemma 2 given value matching $V(\underline{h})=F(\underline{h})$ and smooth pasting $V'(\underline{h})=F'(\underline{h})$, apply Equation (A.3), and solve for $V(\overline{h})-F(\underline{h})-\Delta h$ which is financiers' rents. The inequality follows from $\rho F(\underline{h})-r\underline{h}F'(\underline{h})\geq\mathcal{H}(F)(\underline{h})$ since $\underline{h}\in B\setminus\{0\}$. □

For clarity, first consider startups with $\sigma=0$ such that the lower bound in (12) attains. Lemma 3 states that, in equilibrium, firms that raise financing after running out of funds pay financing costs proportional, by a factor of financiers' bargaining power $1-\theta$, to firm value net of financing $V(\overline{h})-\overline{h}$, whereas firms that raise financing early pay financing costs proportional, by a factor of the ratio of carry cost to the speed of accessing alternative financing $\frac{\varphi}{\gamma}$, to financing amount. Thus, the model predicts what in equilibrium resemble two distinct types of transaction costs – 'fixed' costs covarying with firm value and 'variable' costs proportional to financing amount – as in fact arising from firms' optimal choice of whether to bargain early with financiers for funds.

The reason why the expression $\frac{\varphi}{\gamma}\Delta h$ in (12), given $\sigma=0$, does not involve bargaining power θ is because of early financing. From the discussion of Lemma 2, earlier financing at $h_t>0$ reduces financiers' rents by the amount $(1-\theta)\gamma\big(V(h_t)-V_o(h_t)\big)\,dt$. Next, note that (i) if $h_t\leq\underline{h}$, then $V(h_t)=x(h_t)V(\overline{h})$ by the optimality of bargaining, such that $V(h_t)-V_o(h_t)=x(h_t)V(\overline{h})-V_o(h_t)$ is insiders' surplus from financing at h_t, and (ii) by a standard property of bargaining, $\theta:1-\theta$ equals the ratio of insiders' surplus to financiers' rents, as Equation (3) implies. Therefore, $h_t\leq\underline{h}$ implies that

$$\begin{aligned}(1-\theta)\gamma\underbrace{\big(V(h_t)-V_o(h_t)\big)}_{\text{Reinclusion gains}}&\overset{\text{(i)}}{=}(1-\theta)\gamma\underbrace{\big(x(h_t)V(\overline{h})-V_o(h_t)\big)}_{\text{Insiders' surplus}}\\&\overset{\text{(ii)}}{=}\theta\gamma\underbrace{\Big(\big(1-x(h_t)\big)V(\overline{h})-\big(\overline{h}-h_t\big)\Big)}_{\text{Financiers' rents}}.\end{aligned}$$

Thus, earlier financing at $h_t\in(0,\overline{h}]$ reduces financiers' rents by a *factor* of $\theta\gamma\,dt$. And it involves a carry cost of $\varphi(\overline{h}-h_t)\,dt$, but insiders bear only a fraction θ of it due to bargaining. Therefore, at an optimal 'interior' financing threshold $\underline{h}>0$ that equalizes the marginal costs and benefits of earlier financing, θ is canceled out, giving (12).

Economically, the cancellation of θ reflects that by financing early, insiders are in effect offsetting financier bargaining power $1 - \theta$ by bargaining with financial slack $\underline{h}$.

Robustness to option effects. Generally when $\sigma > 0$, there is a third effect from the option value of waiting. When $h_t \to \underline{h}^+$, delaying financing may avert financing if insiders subsquently receive positive cash flows $\mu\, dt + \sigma\, dB_t > 0$; its expected value is $\frac{1}{2}\sigma^2\big(V''^{(+)}(\underline{h}) - V''^{(-)}(\underline{h})\big)$. This pushes down financing threshold and, as (12) shows, adds an option component to financing costs on top of the 'variable-cost' component, because waiting for an option value worsens *outside* options at bargaining if negative cash flow shocks follow and thus endogenously increases financing costs.

The key comparative statics in Section 5.5 is, however, largely robust to this option effect precisely because of early financing. When firms bargain with financial slack, they are reducing the size of financing costs and thereby lowering the option value of avoiding bargaining in the first place. Thus, the size of the effect mostly comoves with overall financing costs that are being optimized through early financing.

5.5 Comparative statics of financial slack

I now present comparative statics. While some claims are formally established only under the analytically tractable case of $\sigma = 0$ without the option effect, numerical simulations suggest that the patterns are robust even when $\sigma > 0$, consistent with the discussion on it in Section 5.4. For clarity, let $r \geq 0$; a negative yield (very) slightly complicates exposition. I will first explain intuitions and then discuss the significance.

Proposition 3 (Comparative statics in θ and γ)**.** *Equilibrium* $(\underline{h}, \overline{h})$ *is invariant to* $\gamma \leq \underline{\gamma}$. ***Funding target*** $\overline{h}$ *is decreasing in* θ*, and decreasing in* $\gamma \geq \underline{\gamma}$.[28] ***Financial slack vanishes,*** $\overline{h}$, $\underline{h}$, $\Delta h \to 0$*, as either* $\theta \to 1$ *or* $\gamma \to +\infty$. ***Financing threshold*** $\underline{h}$ *is zero for any* θ *above some* $\underline{\theta} < 1$*, and positive and non-monotonic for* $\gamma \in (\underline{\gamma}, \infty)$.

If $\sigma = 0$*, the following also hold:* ***Financing threshold*** $\underline{h}$ *is decreasing in* θ *when* $\underline{h} > 0$*;* ***Financing amount*** Δh *– in case financing is early (i.e.,* $\underline{h} > 0$ *such that* $\Delta h < \overline{h}$*) – is constant (increasing) in* θ *if* $r = 0$ *(*$r > 0$*), and decreasing in* γ *if* $r = 0$*.*

Intuitions. First, an increase in either bargaining weight θ or access to alternative financing γ ($\geq \underline{\gamma}$) lowers incentives to keep financial slack, and hence decreases funding target $\overline{h}$ – down to zero in the limit along with $\underline{h}$ and Δh.[29] Second, funding reserve

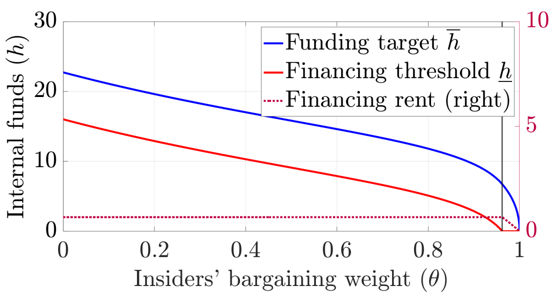
[28] Within this proposition, 'decreasing'/'increasing' indicates strict monotonicity only.

[29] If $\theta = 0$, any allocation is an equilibrium since insiders are indifferent. Thus, the game is not upper-hemicontinuous as $\theta \to 0$. Note that the surplus from financing is *fully* 'externalized' to financiers when $\theta = 0$. Thus, there exists a first-best equilibrium where insiders with choice adopt the best strategy for financiers with benefit. Still, due to full dilution, insiders' equilibrium value function is given by lack of access to any financing, i.e., coinciding with V_o given $\gamma = 0$.

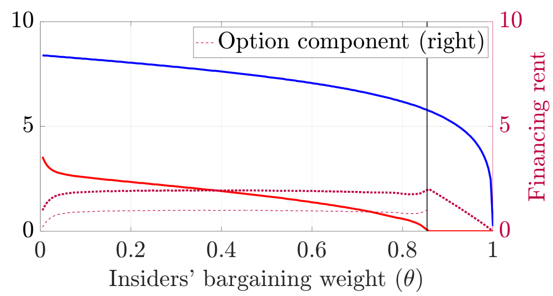
$\underline{h}$ is decreasing in θ as well. As Lemma 3 shows, insiders reduce the size of dilution by financing early $\underline{h} > 0$. With a higher θ, financiers extract a smaller rent, and so insiders have less incentives to reduce the size of dilution.

Funding reserve $\underline{h}$ is, however, non-monotonic in access to alternative financing γ. With γ above but near $\underline{\gamma}$, insiders finance early to improve their outside options. It still takes quite long to find another financier, so that large funding reserves are needed to make it sufficiently likely before the business fails. As alternative financiers become quicker to find, insiders need less funding reserves $\underline{h}$ to make it at least as likely.

Lastly, financing amount Δh determines the frequency of financing, and hence of dilution. Lemma 3 shows that given $\sigma = 0$ and early financing, the size of dilution is $\frac{\varphi}{\gamma}\Delta h$, which is constant in θ and decreasing in γ. Therefore, its optimal frequency, which is decreasing in financing amount Δh, is constant in θ and increasing in γ.[30]



(a) Startups



(b) Operating firms

Figure 6: Comparative statics in bargaining power θ

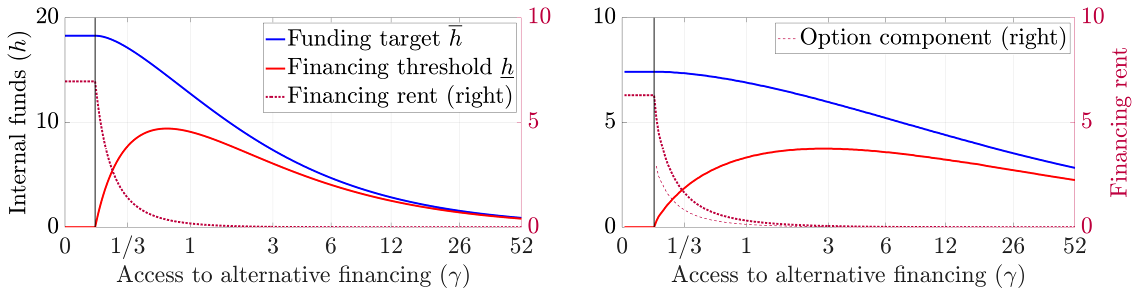
Parameters are $(\rho, r, \gamma) = (0.05, 0, 0.5)$ and, for startups, $(\kappa, \lambda, \Pi) = (2, 0.1, 50)$ and, for operating firms, $(\mu, \sigma) = (1, 2)$; see Section 4.1. Frictionless net present values are set at 20 for both cases. Vertical lines are early financing thresholds. The 'option component' in Figure 6b is the second term in (12) from Lemma 3, which is zero for startups in Figure 6a since $\sigma = 0$; its displayed pattern when θ is close to zero seems driven by numerical imprecision that arises because $V = V_o$ when $\theta = 0$ such that the term converges to a $\frac{0}{0}$ form.

Significance. Proposition 3 is significant in two regards. First, as Figure 6a shows (and as Figure 6b closely aligns), financing rent is increasing in financiers' bargaining power $1-\theta$ only up to $\underline{h} = 0$: on the part of the domain for θ over which financing threshold is positive $\underline{h} > 0$, financing rent is nonincreasing in financiers' bargaining power $1-\theta$, consistent with the discussion of Lemma 3.[31] This suggests that it may be incorrect to observe small financing costs that some firms incur and then conclude that financiers must have small bargaining power vis-à-vis these firms. The small financing costs in equilibrium may in fact be the result of these firms optimally choosing to

[30]If $r > 0$, a higher $\overline{h}$, holding financing amount Δh fixed, implies less frequent financing due to higher yields $rh_t\,dt$ on $h_t \in (\overline{h} - \Delta h, \overline{h}]$; since $\frac{\partial \overline{h}}{\partial \theta} < 0$, Δh is thus increasing in θ when $\underline{h} > 0$.

[31]While Proposition 3 does not discuss financing rent, its displayed patterns in Figure 6a when financing is early are implied by Lemma 3: given $\sigma = 0$ and conditional on early financing, financing amount Δh is, by Proposition 3, constant (decreasing) in financiers' bargaining power $1-\theta$ given $r = 0$ ($r > 0$), and so is financing rent by Lemma 3.

bargain with financiers having large bargaining power $1-\theta \gg 0$ when they have *correspondingly* large financial slack $\underline{h} \gg 0$. In short, firms may offset financiers' bargaining power by financing early.



(a) Startups

(b) Operating firms

Figure 7: Comparative statics in quality of access to financing γ

$\theta = 0.5$. See Figure 6 for explanation of other parameters and graph components. Financing rent is never zero.

Second, the non-monotonicity of financing threshold $\underline{h}$ in access to alternative financing γ, as shown in Figure 7, suggests broad applicability. Conceptually, varying γ over its domain $[0, \infty)$ covers firms facing perfectly monopolistic financiers up to firms facing perfectly competitive financiers asymptotically. Excluding the extreme right tail close to perfect competition, this model states that there is a spectrum of firms along two distinct types of financing dynamics. On one hand, firms that cannot quickly access alternative sources of financing raise financing highly infrequently in large chunks and only when they have exhausted most or all of their internal funds. On the other hand, firms that can do so quickly raise financing highly frequently in small increments even as they always keep a sizable amount of cheaper internal funds.[32] Many classes of firms – including young startup firms, financially distressed firms, and large, established and creditworthy firms – seem to fall onto different points along this spectrum.

And this broad applicability directly arises from this paper's core insight that, because of the bargaining frictions, the incentive to raise financing early increases in the quality of firms' access to financing. Young startup firms and distressed firms have poor access to alternative financing sources, and therefore raise financing highly infrequently in large chunks and only when funds are exhausted. Established and creditworthy firms have great access to alternative financing sources, and *therefore* raise financing even when they have large financial slack. In both cases, financiers may possess sizable bargaining power. But how much of it translates into observed financing costs depends endogenously on how firms respond to it through their optimal financing strategy given their quality of access to financing.

[32]Consistent with the intuition, Supplemental Appendix SA.4 shows that such firms may have funding reserve $\underline{h}$ that substantially exceeds their investment needs – see Figure SA.3b.

6 Investment Extension

This section considers investment as firms' explicit choice to analyze how it interacts with the bargaining frictions. Firms that can choose how much to invest have some control over the flow of funds. As such, investment choice should affect firms' outside options when they bargain with financiers. Specifically, firms that can quickly obtain large extra funds by divesting capital or reducing their investment expenses have good outside options when they keep funds as they bargain: if external financing were to fail, such firms can do so in order to avoid running out of internal funds. Financiers, therefore, can only extract small rents from such firms when they raise financing early.

In this paper, divestment and reducing investment expenses are thus equivalent off-path responses to reduce on-path dilution. But they are distinct economic activities; the idea of investment irreversibility highlights that divesting existing capital is often difficult – such as for intangible capital or during downturns. Investment choice thus creates two separate channels – divestment and reducing investment expenses – through which this paper's mechanism based on financier bargaining power may operate.

This section explores how firms with investment technology choose optimal financing strategy vis-à-vis financiers possessing bargaining power. Section 6.1 extends the setup with investment choice, and Section 6.2 explains how investment affects financing costs depending on its reversibility or lack thereof.

6.1 Extended setup

I modify the cash flow profile with a standard investment model à la Hayashi (1982).

Production and investment. A firm owns capital K_t and employs it to produce

$$\big(A\ dt + \sigma\ dB_t\big)\ K_t$$

in cash flow, with $A, \sigma > 0$. The firm can invest $I_t \in \mathcal{I}$ into its capital stock, subject to adjustment cost; $\mathcal{I} = \mathbb{R}$ means that capital can be divested into funds, whereas $\mathcal{I} = \mathbb{R}_+$ means that it cannot, that is, investment is perfectly irreversible. Given flow investment $I_t\ dt$, the firm incurs an additional flow cost in funds $\Psi(I_t/K_t)K_t\ dt$; Ψ satisfies $\Psi(0) = \Psi'(0) = 0 < \Psi''$. When the firm is neither financing nor paying dividends, internal funds $H_t \geq 0$ evolve as: writing $i_t \equiv I_t/K_t$,

$$dH_t = \Big(A - i_t - \Psi(i_t)\Big)K_t\ dt + \sigma K_t\ dB_t.$$

For simplicity, assume a quadratic cost: $\Psi(i) \equiv \psi\frac{i^2}{2}$ for some $\psi > 0$. Given depreciation rate $\delta \geq 0$, the firm's capital stock K_t evolves as $dK_t/K_t = \big(i_t - \delta\big)\ dt$. While there is no explicit capital trade, $i_t < 0$ given $\mathcal{I} = \mathbb{R}$ means divestment of capital into funds.

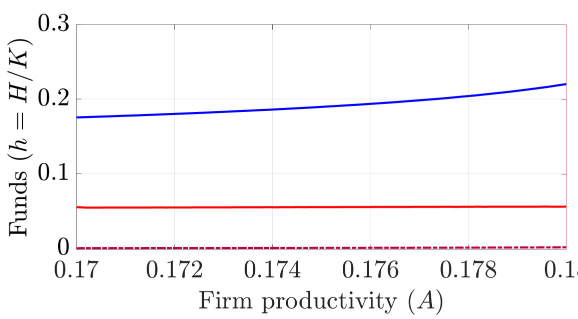
Investment optimization. Define $W(K,H)$ as firm value given capital $K > 0$ and funds $H \geq 0$. Letting $h \equiv H/K$ and $V(h) \equiv W(1,h)$, homogeneity in (K,H) gives $W(K,H) = KV(h)$. Per-capital HJB equation is

$$\rho V(h) - rhV'(h) = \max_{i \in \mathcal{I}} \left\{ \big(A - i - \Psi(i)\big)V'(h) + \frac{1}{2}\sigma^2 V''(h) + \big(i - \delta\big) \underbrace{\big(V(h) - hV'(h)\big)}_{=W_K} \right\}.$$

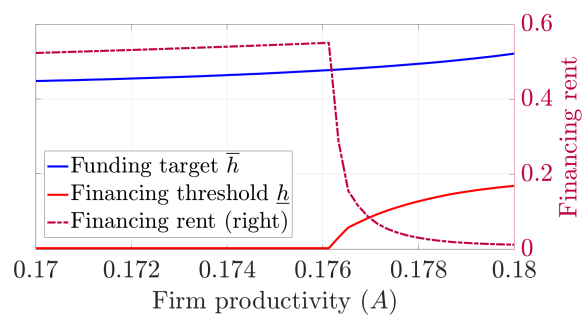
No alternative financiers. To highlight how investment choice affects financing strategy, I set $\gamma = 0$ in this extension. This is also sensible given the present focus on investment irreversibility: finding alternative financiers upon a failure of financing is likely more difficult when either the firm's capital is largely intangible and thus hard to liquidate or the economy is in a downturn such that capital is illiquid in general.

6.2 Early financing and investment irreversibility

Because investment choice limits analytic tractability, here I focus on delivering the intuition. The key goal of the exposition is to show that the specific numerical patterns to be presented arise directly from the same economic mechanism analyzed in Sections 3 through 5. Appendix A.5 provides suggestive analysis in support of the explanation.



(a) Reversible investment $i \in \mathbb{R}$



(b) Irreversible investment $i \geq 0$

Figure 8: Investment irreversibility, financing strategy, and financing costs

Each plot is comparative statics in firm productivity $A \in [0.17, 0.18]$, depending on reversibility of investment. Bargaining parameters are $(\theta, \gamma) = (0.5, 0)$, and other parameters (except for A being varied) are adopted from Bolton et al. (2011): $(\rho, r) = (0.06, 0.05)$, $(A, \sigma, \psi, \delta) = (0.18, 0.09, 1.5, 0.1007)$. Both y-axes are identical across plots, and on a per-capital basis. In both plots, financing rent is never exactly zero.

Figure 8 depicts optimal financing strategy as a function of firm productivity, depending on whether investment is irreversible. First, consider the case of reversibility $i \in \mathbb{R}$ in Figure 8a. Despite the absence of alternative financiers upon a bargaining failure $\gamma = 0$, firms over the domain $A \in [0.17, 0.18]$ still raise financing early, and financing costs are small. But in the case of irreversible investment $i \geq 0$ in Figure

8b, it is only more productive firms that raise financing early, whereas less productive ones delay financing until funds are depleted and incur sizable financing costs.

This stark difference in firms' cross-section of financing costs depending on irreversibility stems from this paper's core claim: financing costs are endogenous to firms' financing strategy, because their outside options at bargaining are endogenous to it. To make the point, I proceed in two steps: I first (i) illustrate how firms' ability to reduce investment incentivizes early financing, and then (ii) explain how irreversibility of investment differentially affects this incentive based on productivity.

(i) Underinvestment as a strategic choice. Suppose the firm has failed in bargaining with financiers. Since $\gamma = 0$ in the present setup, it cannot raise external financing ever again. But if it can, for example, sell off its existing capital stock, the firm can occasionally do so in response to a negative shock in order to avoid business failure. Therefore, firms that can *strategically* underinvest in response to a bargaining failure have good outside options vis-à-vis financiers.

In this context, early financing improves firms' outside options if it takes time for underinvestment to generate sufficient funds. The convex adjustment cost Ψ in the present setup implies that divestment must be optimally smoothed out over time. This captures the idea that disposal of existing capital stock typically requires that the firm needs to find a willing buyer, which may be more difficult for a larger chunk of capital. Accordingly, firms that have lost access to external financing can divest smoothly upon negative shocks if they are sufficiently away from the brink of depleting their funds.

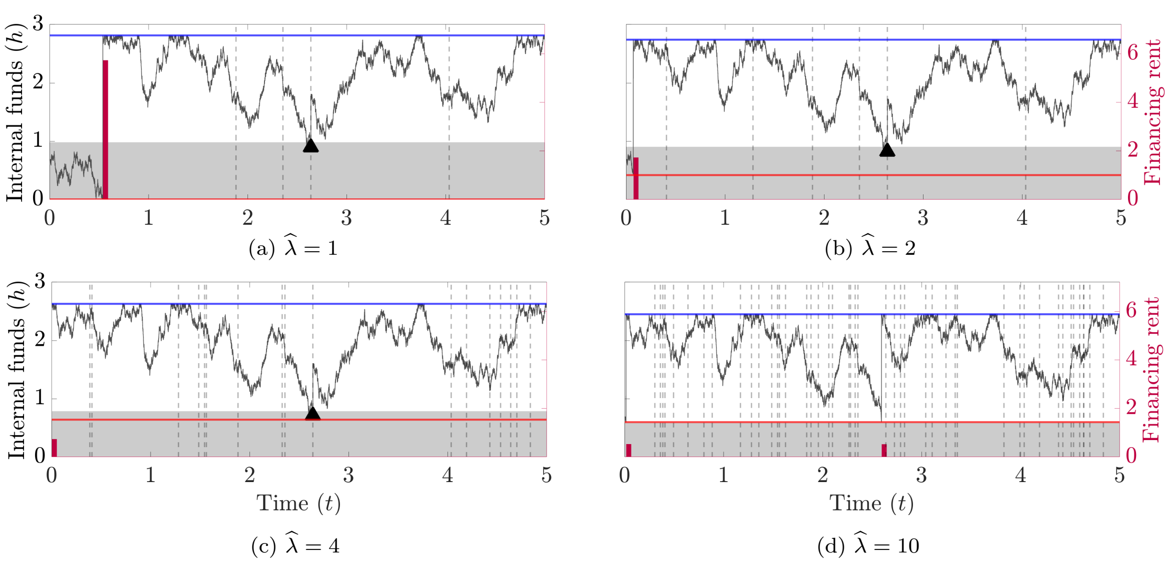


Figure 9: Ease of divestment opportunities $\widehat{\lambda}$, given $\gamma = 0$

Plots track internal funds, dilution and divestment given stochastic arrivals of divestment opportunities (vertical dashed lines in gray). In the gray area above financing threshold in red, divestment is optimal on the path of equilibrium; the black marker indicates actual divestment. All axes are identical across plots.

To illustrate, I modify the main model from Section 4 with lumpy divestment opportunities. The business has 'normalized' running cash flow $\pi \, dt + \sigma \, dB_t$, similar to operating firms. At a Poisson arrival rate of $\widehat{\lambda} > 0$, the business obtains opportunities to downsize future cash flow by a factor of $\eta \in (0,1)$ in return for receiving funds $\xi > 0$. I let $(\rho, r) = (0.07, 0)$, $(\pi, \sigma) = (1, 1)$ and $(\eta, \xi) = (0.9, 0.7)$ so that divestment is not first-best. Lastly, $\theta = 0.5$ and there are no alternative financiers $\gamma = 0$.

Consistent with the above discussion, Figure 9 shows that firms that can divest more swiftly can improve their outside options vis-à-vis financiers when they raise financing with financial slack. This is because if bargaining were to fail, firms with a higher $\widehat{\lambda}$ have a better chance for timely divestment given that they have some funds to begin with. These firms thus choose to raise financing early because they are better at reducing the size of financing costs by improving their outside options.

The case of $\widehat{\lambda} = 10$ in Figure 9d is particularly enlightening. Because financing costs are (endogenously) so low, these firms never divest in equilibrium – the gray area where divestment is optimal is below the financing threshold in red. Thus, their equilibrium dynamics is indistinguishable from operating firms in Sections 4 and 5. The fact that early financing still arises despite the absence of alternative external financing $\gamma = 0$ shows that this *off-path* ability to underinvest is, in a sense, a form of alternative *self*-financing – that is, a substitute for γ.

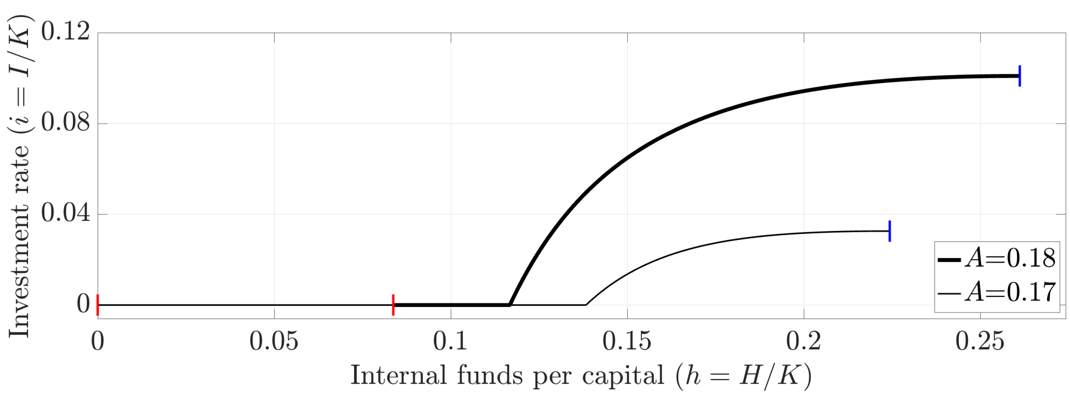


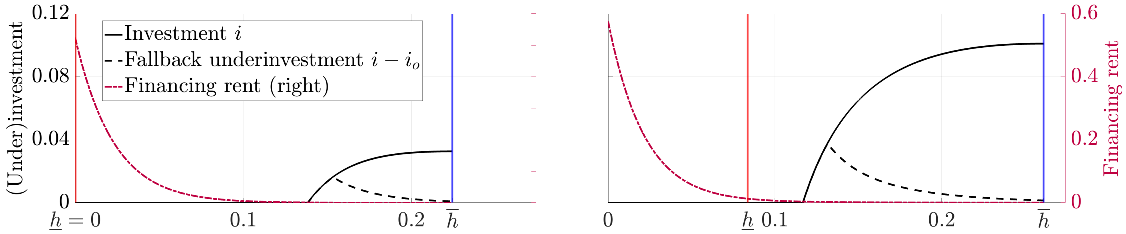
Figure 10: No alternative external financiers $\gamma = 0$ and no divestment $i \geq 0$

The black curves, in different thickness, represent optimal investment policy given internal funds on the equilibrium path, bounded by funding target $\overline{h}$ in the blue line segments and financing threshold $\underline{h}$ in the red ones.

(ii) Heterogeneous effects of investment irreversibility. Now I return to the case with convex investment adjustment cost. Based on the above discussion, the part of Figure 8b given investment irreversibility where financing is not early $\underline{h} = 0$ makes sense: there are no alternative external financiers $\gamma = 0$, and divestment is excluded – both on and off equilibrium path. Nevertheless, firms with higher productivity still raise financing early. To understand why productivity matters for the early financing incentive under irreversibility, I begin by discussing investment policy.

Figure 10 compares two cases of $A \in \{0.17, 0.18\}$, depicting optimal investment rate on the equilibrium path $i(h)$ as a function of internal funds per capital h bounded by funding target $\overline{h}$ and financing threshold $\underline{h}$. As shown, one obvious effect of productivity A is that more productive firms invest more.

The reason why only productive firms raise financing early when investment is irreversible is exactly because this differential on-path investment has crucial off-path implications. Specifically, more productive firms that invest more on path have greater room to cut down on expenses to save funds *if* necessary. In contrast, less productive firms invest less on path, thereby endogenously constraining their ability to reduce investment off path, given that divestment is precluded. Thus, more productive firms have greater incentives to raise financing early in order to improve their outside options and reduce financing costs when bargaining with financiers.



(a) Low productivity $A = 0.17$

(b) High productivity $A = 0.18$

Figure 11: Investment and fallback underinvestment

Fallback underinvestment $i(h) - i_o(h)$ is how much firms with funds h would reduce investment upon losing access to external financing, e.g., due to a failure of bargaining (off the equilibrium path). Both y-axes are identical across plots. All values are on a per-capital basis. In both plots, financing rent is never exactly zero.

Figure 11 illustrates the mechanism. Each subplot contains the investment policy in Figure 10 for each productivity case and additionally depicts two objects. First, 'fallback underinvestment' is the difference between firms' optimal investment depending on whether they have access to financing $i(h)$ or do not $i_o(h)$. Second, 'financing rent' is the amount that the firm must pay financiers if they choose to raise financing given funds h – which is thus suboptimal for $h > \underline{h}$.

The key observation is that due to investment irreversibility, productive firms can underinvest more if bargaining were to fail. Because on-path investment $i(h)$ is higher in Figure 11b, fallback underinvestment $i(h) - i_o(h)$ is higher than in Figure 11a.[33] As such, productive firms raise financing early to improve their outside options with their endogenous ability to underinvest, whereas unproductive firms delay financing until funds are depleted and incur endogenously large financing costs.

[33] With reversibility, unproductive firms may increase fallback underinvestment via $i_o(h) < 0$.

In sum, investment irreversibility disproportionately increases financing costs for unproductive firms, because productive firms invest heavily and therefore can have good outside options if financing were to fail. Thus, this paper generates rich cross-sectional implications on firm financing in environments where capital is difficult to liquidate, such as for intangible capital or during business downturns. And the result directly arises from the model's core economic mechanism that because of financier bargaining power, it is only firms with good financing alternatives – in the present case, the ability to reduce investment expenses – that choose to raise financing early in order to improve their outside options and thereby reduce financing costs.

7 Conclusion

I present a dynamic theory of firm financing based on financier bargaining power. It predicts that firms choose to raise financing in a lumpy fashion and maintain internal funds in order to bargain with financiers – and pay rents – infrequently. Moreover, firms may choose to raise financing early, before running out of internal funds, in order to bargain when their outside options are better due to the ability to pursue financing alternatives. Thus, lumpy financing reduces the frequency of financiers' rent extraction and early financing its size. In short, it is how firms use financial slack to improve their bargaining with financiers that determines the dynamics of firm financing.

The paper provides a new theory of economic inaction based on the optimality of infrequent bargaining, as an alternative to the canonical paradigm based on fixed transaction costs dating to Allais (1947), Baumol (1952) and Tobin (1956). Furthermore, the model endogenizes financing costs such that they not only determine firms' optimal financing strategy, but also are determined by it. This feature leads to a unique prediction that firms with good access to financing face small financing costs in equilibrium because they choose to raise financing early before exhausting their internal funds. In contrast, firms with limited financing alternatives are unable to reduce the costs as much and instead choose to delay financing until internal funds are depleted, thereby incurring large financing costs in equilibrium.

The model's key driver is the idea that if bargaining were to fail, it takes time to access alternative sources of funding. In addition to the main analysis, I extend this idea in the context of investment choice subject to adjustment frictions. The extended model suggests that investment irreversibility not just increases financing costs in general, but disproportionately so for less productive firms that invest less.

Looking forward, this paper suggests further research in three broad directions. First, it generates a cross-sectional prediction that financial slack is non-monotonic and may be locally decreasing in how specialized firms' financing relationships are – such as startups tied to stage-specific venture capital or borrowers tied to a single lead arranger in syndicated lending. This can be potentially tested using empirical proxies such as lender concentration, relationship tenure, or, simply, firms' credit ratings. Second, the model's restriction to equity financing raises in turn a theoretical question of optimal dynamic security design in the presence of financier bargaining power. Third, the implication from investment irreversibility suggests that financier bargaining power may contribute to widening cross-sectional dispersion in firm growth in intangible-intensive sectors or during business downturns.

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Appendix A Omitted Proofs And Analysis

A.1 Lemma 1 (Existence and uniqueness).

Proof. Let $\mathcal{W}$ be the space of continuous nonnegative bounded functions on $\mathbf{H} \equiv [0, H]$ with an arbitrary $H \gg 0$, and complete it with L^∞ supremum metric. Keep cash flow as in Section 4.1, and also the assumptions on dividends and business failure upon funding depletion without prompt financing.[34] Define $T, T_o : \mathcal{W} \to \mathcal{W}$ as: for $W \in \mathcal{W}$,

1. $h \mapsto T_o W(h)$ is the value function of 'excluded' insiders lacking the ability to raise financing (so that the boundary condition is $T_o W(0) = 0$), who, upon 're-inclusion' that occurs at the stopping time τ_γ given by a Poisson rate $\gamma \in [0, \infty)$, receive a terminal payoff of $W(h_{\tau_\gamma})$; and

2. $h \mapsto TW(h)$ is the value function of non-excluded insiders who can, at any $t \geq 0$, either (i) forgo financing, or (ii) raise financing through bargaining with financiers where insiders' outside option from bargaining is given by $W(h_t)$:

   - if $W(h_t)$ is sufficiently high, 'bargaining' involves zero financing and zero rents since (iii) insiders *take* the outside option such that $TW(h_t) = W(h_t)$.

By the Theorem of the Maximum, the above transformations are well-defined as self-maps given the continuity restriction in $\mathcal{W}$.

An equilibrium value function V is a fixed point of the *concatenated* map $\widehat{T} \equiv T \circ T_o$, and vice versa; if V exists, then $V_o \equiv T_o V$. If $\widehat{T}$ is a contraction on $\mathcal{W}$, by contraction mapping theorem there exists a unique fixed point $V \in \mathcal{W}$ such that $V = \widehat{T}V$.

The claim on $\widehat{T}$ follows from Blackwell's Lemma. First, obviously $\widehat{T}$ is nondecreasing: $W' \geq W$ on $\mathbf{H}$ implies $\widehat{T}W' \geq \widehat{T}W$ on $\mathbf{H}$. Second, let us show that there exists some $\zeta \in [0, 1)$ such that for any $W \in \mathcal{W}$ and $w \in \mathbb{R}_{++}$, $\widehat{T}(W + w) \leq \widehat{T}W + \zeta w$.

Recall that τ_γ occurs at a Poisson rate $\gamma \in [0, \infty)$. Then,

$$T_o(W + w) \leq T_o W + \zeta_\gamma^\rho w, \tag{A.1}$$

where $\zeta_\gamma^\rho \equiv \mathbb{E}\left[e^{-\rho\tau_\gamma}\right] = \frac{\gamma}{\gamma+\rho} < 1$.[35] As an aside, T_o is a contraction on $\mathcal{W}$, and its unique fixed point is invariant to $\gamma \in [0, \infty)$. Next, for any $w \in \mathbb{R}_{++}$,

$$T(W + w) \leq TW + w. \tag{A.2}$$

[34] Also enforce that whenever $h_t > H$, dividend payout of at least $h_t - H$ is mandatory.

[35] The Poisson structure is not essential; one can, for example, assume that re-inclusion deterministically occurs at $\tau_\gamma \equiv 1/\gamma > 0$, in which case $T_o W(h)$ is excluded insiders' value at $t = 0$ with $h_0 = h$ given the otherwise identical environment. As long as re-inclusion takes some time with nonzero probability $\mathbb{P}(\tau_\gamma > 0) > 0$, the bonus dividend $w \in \mathbb{R}_{++}$ that is received only upon re-inclusion is, from $t = 0$, discounted in expectation (at least) by a factor of $\mathbb{E}[e^{-\rho\tau_\gamma}] < 1$.

This is because marginal benefit from uniform global improvement in reservation value function does not exceed unity. The upper bound is attained wherever on $h_t \in \mathbf{H}$ (iii) the outside option $W(h_t)$ is so good that insiders take it, $TW(h_t) = W(h_t)$. Otherwise, marginal benefit is $1 - \theta$ wherever on $\mathbf{H}$ insiders optimally (ii) raise positive funds through bargaining, and zero wherever on $\mathbf{H}$ insiders optimally (i) forgo financing.[36]

T is not strictly a contraction, but it is not 'expansionary': its contraction coefficient is unity. Therefore, $\widehat{T}$ inherits the coefficient $\zeta_\gamma^\rho \in [0, 1)$ from T_o: for any $w \in \mathbb{R}_{++}$,

$$\widehat{T}(W + w) = T\big(T_o(W + w)\big) \leq T\big(T_oW + \zeta_\gamma^\rho w\big) \leq T\big(T_oW\big) + \zeta_\gamma^\rho w = \widehat{T}W + \zeta_\gamma^\rho w.$$

The first inequality is from Inequality (A.1) and monotonicity of T, and the second from Inequality (A.2). By Blackwell's Lemma, $\widehat{T}$ is a contraction on $\mathcal{W}$, as desired.[37] □

A.2 Lemma 2 (Monotone financing strategy)

Proof. (Continued from Section 5.2) Rewrite Inequality (11) as: for $h \in B \setminus \{0\}$, $G(h) \equiv (1 - \theta)\gamma\big(V(h) - V_o(h)\big) - \theta\varphi\big(\overline{h} - h\big) \geq 0$. Define $\widetilde{G}$ as follows: for $h \in [0, \overline{h}]$,

$$\widetilde{G}(h) \equiv (1 - \theta)\gamma\big(F(h) - V_o(h)\big) - \theta\varphi\big(\overline{h} - h\big).$$

Obviously, $\widetilde{G} \leq G$, and $\widetilde{G}(h) = G(h)$ if and only if $h \in B$. Note that

$$\widetilde{G}'(h) = \theta\Big(\varphi - (1 - \theta)\gamma\big(V_o'(h) - 1\big)\Big),$$

which implies that (i) $\widetilde{G}'' \geq 0$ given weak concavity of V_o from standard dynamic programming arguments, and (ii) $\widetilde{G}'(h) < 0$ if and only if $F'(h) > 1 + \frac{\varphi}{\gamma}$. Also,

$$(1 - \theta)\big(F(h) - V_o(h)\big) = \theta\big(V(\overline{h}) - F(h) - (\overline{h} - h)\big), \tag{A.3}$$

which is an identity that reflects standard properties of bargaining.

I establish an intermediate claim: if $h \in B \setminus \{0\}$, then $\rho F(\widetilde{h}) - r\widetilde{h}F'(\widetilde{h}) > \mathcal{H}(F)(\widetilde{h})$ for any $\widetilde{h} < h$. Since $F(h) = V(h)$ and $\widetilde{G}(h) = G(h) \geq 0$ by $h \in B \setminus \{0\}$, it holds that $\theta\gamma\big(V(\overline{h}) - V(h) - (\overline{h} - h)\big) \geq \theta\varphi(\overline{h} - h)$ by Equation (A.3), and thus,

$$\frac{V(\overline{h}) - V(h)}{\overline{h} - h} = \frac{1}{\overline{h} - h}\int_h^{\overline{h}} V'(y)\, dy \geq 1 + \frac{\varphi}{\gamma}.$$

Since V is weakly concave (by standard arguments) and $V'(\overline{h}) = 1$, it must be that

[36] These are pointwise effects of a rise in $W(h)$ on $TW(h)$. When W globally shifts up uniformly, marginal effect on any $TW(h)$ is a weighted average of 1, $1 - \theta$ and 0, and hence unity at most.
[37] Supplemental Appendix SA.2 graphically illustrates the proof.

$V'(h) > 1 + \frac{\varphi}{\gamma}$. Note that $F'(h) = V'(h)$ either trivially if h is in the interior of B or by smooth pasting otherwise, such that $F'(h) > 1 + \frac{\varphi}{\gamma}$ and so $\widetilde{G}'(h) < 0$. Since $\widetilde{G}'' \geq 0$ and $V \geq F$ globally, it follows that $0 < \widetilde{G} \leq G$ on $[0, h)$, establishing the claim.

Now, suppose by way of contradiction that there exists some $\widehat{h} \in B \setminus \{0\}$ such that for any sufficiently small $\varepsilon > 0$, $\widehat{h} - \varepsilon \notin B$. Take $h' < \widehat{h}$ such that $h' \in B$ but $(h', \widehat{h})$ is disjoint with B; it exists since $0 \in B$. Define $D \equiv V - F$; it satisfies $D(h') = D(\widehat{h}) = 0$ by value matching but $D > 0$ on $(h', \widehat{h})$. Let h^* be the maximizer of D on $(h', \widehat{h})$, such that $D'(h^*) = 0$ and $D''(h^*) \leq 0$. But then,

$$\begin{aligned}
\rho V(h^*) - rh^* V'(h^*) > \rho F(h^*) - rh^* F'(h^*) &> \mathcal{H}(F)(h^*) \\
&= \mu F'(h^*) + \frac{1}{2}\sigma^2 F''(h^*) + \lambda\big(\Pi + h^* - F(h^*)\big) \\
&\geq \mu V'(h^*) + \frac{1}{2}\sigma^2 V''(h^*) + \lambda\big(\Pi + h^* - V(h^*)\big) = \mathcal{H}(V)(h^*)
\end{aligned}$$

since $V(h^*) > F(h^*)$, $V'(h^*) = F'(h^*)$, and $V''(h^*) \leq F''(h^*)$. This contradicts the HJB equation for V which should hold at $h = h^*$. This completes the proof; for subsequent reference, the present reasoning implies that if $G(0) \leq 0$, then $\underline{h} = 0$. □

A.3 Proposition 2 (Monotonicity of early financing)

Proof. Since $V(0) = F(0)$,

$$G(0) = \theta\Big((1-\theta)\gamma\big(V(\overline{h}) - \overline{h}\big) - \varphi\overline{h}\Big). \tag{A.4}$$

Since Part 1 of Appendix A.4 shows that $V(\overline{h}) - \overline{h} > 0$ is nondecreasing in γ while $\overline{h}$ nonincreasing in γ, it follows that $G(0)$ is negative when $\gamma = 0$, strictly increasing in γ, and strictly positive for sufficiently high γ.[38] Also, note that it is impossible that $F'(0) > V'(0)$; if it holds, $F(0) = V(0)$ implies that $F > V$ on a neighborhood above $h = 0$, a contradiction.

Writing the objects as functions of $\gamma \in [0, \infty)$, it holds that $V'(0; \gamma)$, taken as a derivative of V with respect to h at $h = 0$, is constant in γ as long as $\underline{h}(\gamma) = 0$, because $V_o(0; \gamma)$ is identically zero and thus equilibrium value function V is unaffected by γ when $\underline{h}(\gamma) = 0$. In addition, $F'(0; \gamma) = \theta + (1-\theta)V_o'(0; \gamma)$ is strictly increasing in γ and goes to infinity as $\gamma \to \infty$, because for any $h > 0$, $V_o(h; \gamma) > 0$ is strictly increasing in γ even as $V_o(0; \gamma) = 0$ by the boundary condition. Therefore, the following object is well-defined, positive, and finite:

$$\underline{\gamma} \equiv \inf\big\{\gamma \geq 0 \;\big|\; G(0; \gamma) > 0 \text{ and } F'(0; \gamma) = V'(0; \gamma)\big\} \in (0, \infty).$$

[38] Only *strict* monotonicity in Part 1 depends on Proposition 2.

When $\gamma < \underline{\gamma}$, either $G(0;\gamma) \leq 0$ or $F'(0;\gamma) < V'(0;\gamma)$ such that $\underline{h}(\gamma) = 0$. When $\gamma > \underline{\gamma}$, then $\underline{h}(\gamma) > 0$ since otherwise, $F'(0;\gamma) > V'(0;\gamma)$. Lastly, suppose that $\gamma = \underline{\gamma}$. If $G(0;\underline{\gamma}) = 0$, then obviously $\underline{h}(\underline{\gamma}) = 0$. Suppose that $G(0;\underline{\gamma}) > 0$. For $\gamma < \underline{\gamma}$ sufficiently close, $G(0;\gamma) > 0$ and $\underline{h}(\gamma) = 0$. Letting $\gamma \to \underline{\gamma}^-$, the invariant HJB equation must still hold for $V(h;\underline{\gamma})$ at $h = 0$. Since $F'(0;\underline{\gamma}) = V'(0;\underline{\gamma})$, it follows from $G(0;\underline{\gamma}) > 0$ that $\frac{1}{2}\sigma^2 F''(0;\underline{\gamma}) < \frac{1}{2}\sigma^2 V''(0;\underline{\gamma})$. If $\sigma = 0$, this is a contradiction, and so $G(0;\underline{\gamma}) = 0$. Otherwise, $F''(0;\underline{\gamma}) < V''(0;\underline{\gamma})$ such that $F(h;\underline{\gamma}) < V(h;\underline{\gamma})$ for small $h > 0$. In either case, therefore, $\underline{h}(\underline{\gamma}) = 0$. □

A.4 Proposition 3 (Comparative statics in θ and γ)

Proof. Constancy of $(\underline{h}, \overline{h})$ in $\gamma \in [0, \underline{\gamma}]$ is because $\gamma \leq \underline{\gamma}$ implies $V_o(\underline{h}) = V_o(0) = 0$.

Part 1: Funding target $\overline{h}$. Take $\gamma_2 > \gamma_1 \geq \underline{\gamma}$ and consider the equilibrium with $\gamma = \gamma_2$. When insiders bargain with financiers, they choose $\overline{h}_2$ to maximize $V(\overline{h}_2;\gamma_2) - \overline{h}_2$. Suppose that they agree, as a one-shot deviation, to choose $\overline{h}_1$ instead and then mimic the optimal financing strategy under $\gamma = \gamma_1$ (i.e., refinance at $\underline{h}_1$, pay out above $\overline{h}_1$) until next financing. Denote the payoff function associated with this strategy as $\widetilde{V}$. Note that $\widetilde{V}(\overline{h}_1;\gamma_2) > V(\overline{h}_1;\gamma_1)$, as the reservation value at $h_t = \underline{h}_1 > 0$ is strictly higher with $\gamma = \gamma_2 > \gamma_1$. Since $\overline{h}_2$ without the one-shot deviation is optimal, $V(\overline{h}_2;\gamma_2) - \overline{h}_2 \geq \widetilde{V}(\overline{h}_1;\gamma_2) - \overline{h}_1 > V(\overline{h}_1;\gamma_1) - \overline{h}_1$. Finally, since

$$V(\overline{h};\gamma) - \overline{h} = \frac{1}{\rho + \lambda}\Big(\mu + \lambda\Pi - \varphi\overline{h}\Big), \tag{A.5}$$

from evaluating Equation (9) given smooth pasting $V'(\overline{h}) = 1$ and super contact $\sigma > 0 \implies V''(\overline{h}) = 0$, we have $\overline{h}_2 < \overline{h}_1$. Strict monotonicity in θ is proven similarly.

Part 2: Asymptotics. Since $\overline{h}$ decreases in θ, it converges as $\theta \to 1^-$. Note that

$$V(\overline{h}) - V(\underline{h}) - \Delta h = \int_{\overline{h}-\Delta h}^{\overline{h}} V'(h) - 1 \, dh = (1-\theta)\big(V(\overline{h}) - \Delta h - V_o(\underline{h})\big). \tag{A.6}$$

Since $V(\overline{h}) - \Delta h - V_o(\underline{h}) = V(\overline{h}) - \overline{h} - \big(V_o(\underline{h}) - \underline{h}\big)$ is bounded above in θ by the frictionless value $\frac{\mu+\lambda\Pi}{\rho+\lambda} > 0$, dividend optimality condition $V'(\overline{h}) = 1$ and concavity of V require that $\Delta h \to 0$ as $\theta \to 1^-$. Now, suppose by way of contradiction that $\overline{h} \to \widetilde{h} > 0$ as $\theta \to 1^-$. Note that since $V(\overline{h}) - \overline{h}$ is also bounded above, $G(0)$ as defined in Equation (A.4) converges to $-\varphi\widetilde{h} < 0$. Therefore, $\underline{h} = 0$ for θ sufficiently high by Appendix A.2.[39] But then $\overline{h} = \Delta h \to 0$ as $\theta \to 1^-$, a contradiction.

Next, $\overline{h}$ also converges as $\gamma \to \infty$, again by monotone convergence. Note that $V(h) - V_o(h) \to 0$ as $\gamma \to \infty$ pointwise for $h > 0$, even as $V(0) - V_o(0) = V(0)$ does not

[39] While the statement is true in itself, the actual proof in Part 3 is given $\overline{h} \to 0$ as $\theta \to 1^-$.

converge to zero. Again, suppose by way of contradiction that $\overline{h} \to \widetilde{h} > 0$ as $\gamma \to \infty$. Then, either (i) $\Delta h \to \widetilde{\Delta h} > 0$, or (ii) $\underline{h} \to \widetilde{\underline{h}} > 0$.[40]

In case (i), financiers' rents converge to

$$V(\overline{h}) - V(\underline{h}) - \Delta h \to \int_{\widetilde{h}-\widetilde{\Delta h}}^{\widetilde{h}} V'(h) - 1 \; dh,$$

which is strictly positive by the HJB equation with smooth pasting at $\widetilde{h} > \widetilde{h} - \widetilde{\Delta h}$. Since financiers' rents also equal

$$\left(\frac{1-\theta}{\theta}\right)\big(V(\underline{h}) - V_o(\underline{h})\big),$$

the gap $V(\underline{h}) - V_o(\underline{h})$ does not vanish, which requires that $\underline{h} \to 0$. But any positive limit for financing threshold $\underline{h} \to \varepsilon$ for $\varepsilon \in (0, \widetilde{\Delta h})$, as a deviation, vanishes the gap when $\gamma \to \infty$, and hence the rents that financiers can extract. Thus, the marginal benefit given $\varepsilon \to 0^+$, in terms of reducing financiers' rents, diverges to infinity as $\gamma \to \infty$.[41] In comparison, the marginal costs include additional flow carry cost $\varphi \, dt$ and a higher frequency of financing, and both are bounded. Therefore, there is a strictly profitable deviation for sufficiently high γ, a contradiction.

In case (ii), note that $V(h) - V_o(h)$ converges uniformly to zero on, say, $[\widetilde{\underline{h}}/2, \widetilde{\underline{h}}]$. Therefore, the increase in financing rents when $\underline{h} \to \widetilde{\underline{h}} - \varepsilon$ for $\varepsilon \in (0, \widetilde{\underline{h}}/2)$ vanishes as $\gamma \to \infty$. But this marginally lower financing threshold when $\varepsilon \to 0^+$ has non-vanishing marginal benefits of less flow carry cost $\varphi \, dt$ and lower frequency of financing. As such, there is a strictly profitable deviation for sufficiently high γ, a contradiction.

Part 3: Financing threshold $\underline{h}$. For the first claim on $\underline{\theta}$, I first establish an intermediate claim: $\frac{d\overline{h}}{d\theta} \to -\infty$ as $\theta \to 1^-$. For this, reformulate Equation (A.6) using (A.5) as

$$\int_{\underline{h}}^{\overline{h}} V'(h) - 1 \; dh = (1-\theta)\left(\frac{\mu + \lambda\Pi}{\rho + \lambda} - \frac{\varphi}{\rho+\lambda}\Delta h + \frac{r+\lambda}{\rho+\lambda}\underline{h} - V_o(\underline{h})\right)$$

Totally differentiating both sides with respect to θ gives

$$\begin{aligned}\int_{\underline{h}}^{\overline{h}} \frac{d}{d\theta}\big(V'(h) - 1\big) \; dh - \big(V'(\underline{h}) - 1\big)\frac{d}{d\theta}\underline{h} = -\Big(V(\overline{h}) - \overline{h} - (V_o(\underline{h}) - \underline{h})\Big) \\ + (1-\theta)\left(\frac{r+\lambda}{\rho+\lambda}\frac{d}{d\theta}\underline{h} - \frac{\varphi}{\rho+\lambda}\frac{d}{d\theta}\Delta h - \frac{d}{d\theta}V_o(\underline{h})\right).\end{aligned}$$

Letting $\theta \to 1^-$ trivially vanishes the integral term because $\Delta h \to 0$ while $V'(\overline{h}) = 1$.

[40]If the limits do not exist, Bolzano-Weierstrass Theorem ensures the existence of convergence subsequences given $\gamma_n \to \infty$ as $n \to \infty$.

[41]For clarity, the statement concerns the limit of the derivative $\frac{d}{dh}\big(V(0) - V_o(0)\big)$ as $\gamma \to \infty$.

Rearranging the potentially non-vanishing terms gives

$$\Big(V(\overline{h}) - \overline{h} - \big(V_o(\underline{h}) - \underline{h}\big)\Big) + (1-\theta)\frac{d}{d\theta}V_o(\underline{h})$$
$$= \left(\big(V'(\underline{h}) - 1\big) + (1-\theta)\frac{r+\lambda}{\rho+\lambda}\right)\frac{d}{d\theta}\underline{h} - (1-\theta)\frac{\varphi}{\rho+\lambda}\frac{d}{d\theta}\Delta h + o(1-\theta).$$

The left-hand side is positive and, especially, the first term does not vanish in the limit since $\gamma < \infty$. By Part 2, $\lim_{\theta\to 1^-}\frac{d}{d\theta}\underline{h} \in [-\infty, 0]$. Therefore, it must be that $\lim_{\theta\to 1^-}\frac{d}{d\theta}\Delta h = -\infty$, and thus,

$$\lim_{\theta\to 1^-}\frac{d}{d\theta}\overline{h} = \lim_{\theta\to 1^-}\frac{d}{d\theta}\underline{h} + \lim_{\theta\to 1^-}\frac{d}{d\theta}\Delta h = -\infty.$$

The main claim is straightforward. Note that by Part 1, $G(0)$ from Equation (A.4) goes to zero as $\theta \to 1^-$. Since the argument only concerns the sign of $G(0)$ for θ near 1, divide $G(0)$ by θ and then substitute Equation (A.5). Differentiate the resulting expression with respect to θ to obtain

$$\frac{d}{d\theta}\frac{G(0)}{\theta} = -\gamma\frac{\mu+\lambda\Pi}{\rho+\lambda} - \varphi\left(1 + \frac{(1-\theta)\gamma}{\rho+\lambda}\right)\frac{d}{d\theta}\overline{h}.$$

By the intermediate claim, therefore, $\frac{d}{d\theta}\frac{G(0)}{\theta} \to \infty$ as $\theta \to 1^-$, such that $G(0)$ is strictly negative for $\theta < 1$ sufficiently close. The claim then follows from Appendix A.2.

Next, since (i) $\underline{h}(\underline{\gamma}) = 0$, (ii) $\underline{h} > 0$ for $\gamma > \underline{\gamma}$, and yet (iii) $\underline{h} \to 0$ as $\gamma \to \infty$, non-monotonicity in $\gamma \in (\underline{\gamma}, \infty)$ is proven sufficiently if $\underline{h}$ is continuous at $\gamma = \underline{\gamma}$. Suppose by way of contradiction that $\lim_{\gamma\to\underline{\gamma}^+}\underline{h} = \widetilde{\underline{h}} > 0$.[42] As $\gamma \to \underline{\gamma}^+$, $V = F$ on $[0, \widetilde{\underline{h}}]$ and so $V = F$ fails to satisfy its own HJB on the interval. But at exactly $\gamma = \underline{h}$, V must still satisfy its own HJB on the interval, and hence F must also satisfy that same HJB, such that, still $V = F$ on $[0, \widetilde{\underline{h}}]$. But then $\underline{h}(\underline{\gamma}) = \widetilde{\underline{h}} > 0$, contradicting Proposition 2.

Lastly, strict monotonicity of $\underline{h} = \overline{h} - \Delta h > 0$ in θ given $\sigma = 0$ is immediate from decreasing $\overline{h}$ in Part 1 and nondecreasing Δh in Part 4.

Part 4: Financing amount Δh. Suppose that $\underline{h} > 0$. Rearranging (12) which equals $V(\overline{h}) - V(\underline{h}) - \Delta h$ gives

$$1 + \frac{\varphi}{\gamma} = \frac{V(\overline{h}) - V(\underline{h})}{\Delta h} = \frac{1}{\Delta h}\int_{\overline{h}-\Delta h}^{\overline{h}} V'(h)\, dh. \tag{A.7}$$

That is, early financing specifies the *average* of the first derivative of V over $[\overline{h} - \Delta h, \overline{h}]$. Recall that dividend optimality stipulates that $V'(\overline{h}) = 1$. Therefore, given $\overline{h} > 0$, one can equivalently think of V as satisfying the HJB equation globally below $\overline{h}$ and Δh

[42] Again, Bolzano-Weierstrass Theorem ensures the existence of a convergent subsequence.

as simply choosing when the first derivative on the interval meets the desired average $1 + \frac{\varphi}{\gamma}$.

Since the first derivative is concerned, $V(\overline{h})$ is irrelevant. Given that $\sigma = 0$ restricts to the startup case by Assumption 1, differentiate the HJB using the startup notation in Section 4.1:

$$V''(h) = -\frac{1}{\kappa - rh}\Big[(\varphi + \lambda)V'(h) - \lambda\Big].$$

If $r = 0$, the above equation that determines the evolution of V' below $\overline{h}$ is independent of (θ, γ). Since the desired average is $1 + \frac{\varphi}{\gamma}$, the claims when $r = 0$ are established.

Suppose $r > 0$. Because $V'(h) > 1$ for $h < \overline{h}$ and $\kappa > r\overline{h}$ by Assumption 1, a higher $\overline{h}$ means that $V''(\overline{h} - \delta h)$ is pointwise lower. Therefore,

$$V'(\overline{h} - \delta h) = 1 - \int_{\overline{h}-\delta h}^{\overline{h}} V''(h)\ dh$$

is pointwise higher, and so a higher $\overline{h}$, ceteris paribus, reduces Δh in equilibrium. The claim then follows from Part 1. □

A.5 Supporting analysis on Section 6.2

Fully characterizing optimal financing policy with investment choice is analytically intractable. Here, I conduct suggestive analysis in support of the intuitions offered in Section 6.2. Specifically, I focus on a key necessary condition for the optimality of financing marginally earlier before running out of funds and how investment irreversibility differentially affects it depending on productivity.

As in Section 5.2, define F as the payoff function when insiders immediately raise financing as a one-shot and then revert to the optimal policy. With funds $h = h_t \to 0^+$, financing immediately is preferred to delaying financing by a dt instant (without considering the option value of averting financing) if and only if, omitting h,

$$\rho F - rhF' \geq \big(A - i_F - \Psi(i_F)\big)F' + \frac{1}{2}\sigma^2 F'' + (i_F - \delta)\big(F - hF'\big), \tag{A.8}$$

where i_F is optimal investment policy during the instantaneous delay being considered, given by: if investment is reversible $\mathcal{I} = \mathbb{R}$,

$$i_F = \frac{1}{\psi}\left(\frac{F}{F'} - h - 1\right),$$

and otherwise $\mathcal{I} = \mathbb{R}_+$, the greater of the above and zero.

The same derivation as in the proof of Lemma 2 leads to the following equivalent reformulations of Inequality (A.8) evaluated at $h \to 0^+$. First, in case of reversibility,

$$G_0 \equiv \theta \left(\frac{1}{2\psi} \Big(V(\overline{h}) - \overline{h} - 1 - \psi i_F(0) \Big) \Big(V(\overline{h}) - \overline{h} \Big) - \varphi \overline{h} \right) \geq 0. \tag{A.9}$$

Above, typically $i_F(0) < 0$. To see this, observe that

$$i_F(0) = \frac{1}{\psi} \left(\frac{\theta(V(\overline{h}) - \overline{h})}{\theta + (1-\theta)V_o'(0)} - 1 \right). \tag{A.10}$$

Here, the fraction inside the parenthesis is mostly less than unity and typically close to zero, because $V_o'(0)$ is marginal value of funds at the brink of business failure and so tends to be large due to the extreme curvature.[43] Thus, the incentive for early financing given reversible investment, as captured by G_0, is strengthened by the extent to which firms without access to financing would be incentivized to preserve funds $V_o'(0)$ through divestment.

Second, when investment is irreversible, the above discussion implies that typically, $i_F(0) = 0$. Thus, if $i(\overline{h}) = 0$, then $\widehat{G}_0 = -\theta\varphi\overline{h}$, and otherwise,

$$\widehat{G}_0 \equiv \theta \left(\frac{1}{2\psi} \Big(V(\overline{h}) - \overline{h} - 1 \Big)^2 - \varphi \overline{h} \right) \geq 0. \tag{A.11}$$

Before the main analysis, I first state a handy result.

Lemma A.5.1. *Given that the irreversibility constraint is either absent or slack at funding target, net firm value at funding target is given by*

$$V(\overline{h}) - \overline{h} = 1 + \psi \left((\rho + \delta) - \sqrt{(\rho+\delta)^2 - \frac{2}{\psi}\left(A - \rho - \delta - \varphi \overline{h} \right)} \right). \tag{A.12}$$

Proof. Rearrange the main HJB using $i = \frac{1}{\psi}\left(\frac{V}{V'} - h - 1\right)$ to obtain

$$\begin{aligned} &\left(\rho + \delta + \frac{1}{\psi} \right) V - \left(A + \frac{1}{2\psi} + \left(r + \delta + \frac{1}{\psi} \right) h \right) V' - \frac{1}{2\psi} \left(\frac{V^2}{V'} - 2hV - h^2 V' \right) \\ &- \frac{1}{2}\sigma^2 V'' \equiv \widetilde{\rho} V - \Big(\widetilde{\mu} + \widetilde{r} h \Big) V' - \frac{1}{2\psi} \left(\frac{V^2}{V'} - 2hV - h^2 V' \right) - \frac{1}{2}\sigma^2 V'' = 0. \end{aligned}$$

Rearrange for $\overline{V} \equiv V(\overline{h})$ using smooth pasting and super contact:

$$\Omega(\overline{V}|\overline{h}) = 0,$$

where $\Omega(v|h) \equiv \frac{1}{2\psi}v^2 - \left(\widetilde{\rho} + \frac{h}{\psi} \right) v + \left(\widetilde{\mu} + \widetilde{r}h - \frac{h^2}{2\psi} \right)$. Then, $\Omega_v(\overline{V}|\overline{h}) < 0$, because,

[43] The fraction may still exceed unity if θ is close to unity, but then this paper is not relevant.

letting V^* first-best value, both roots solving $\Omega(v^*|0) = 0$ are strictly between $v^- - \overline{h}$ and $v^+ - \overline{h}$, where $v^\pm$ are the two roots of $\Omega(v|\overline{h}) = 0$, even as $\overline{V} - \overline{h} < V^*$; i.e., $\overline{V} = v^-$.[44] A quadratic formula then gives Equation (A.12). $\square$

Early financing when investment is reversible. Substituting Equation (A.10) reformulates Inequality (A.9) as

$$\frac{G_0}{\theta} = \frac{1}{2\psi}\Big(V(\overline{h}) - \overline{h}\Big)^2 \left(\frac{(1-\theta)V_o'(0)}{\theta + (1-\theta)V_o'(0)}\right) - \varphi\overline{h} \geq 0. \tag{A.13}$$

Under the empirically relevant case where positive investment sometimes occurs in equilibrium, net firm value must strictly exceed unity, since $i(\overline{h}) = \frac{1}{\psi}\big(V(\overline{h}) - \overline{h} - 1\big)$. Therefore, from Lemma A.5.1,

$$\varphi\overline{h} < A - \rho - \delta.$$

Given standard parametrization (see Bolton et al., 2011, for example), the right-hand side above tends to be quite small relative to $\frac{1}{2\psi}$, which is strictly less than $\frac{1}{2\psi}\big(V(\overline{h}) - \overline{h}\big)^2$. Therefore, if $V_o'(0)$ is sufficiently high and θ not too close to unity, Inequality (A.13) holds broadly, which means that when $h_t \to 0^+$, raising financing marginally earlier is mostly preferred to raising financing marginally later.[45]

Early financing under irreversibility. Inequality (A.11) is, in comparison, more difficult to satisfy because now it is net firm value *minus one* that is being squared. In addition, net value should be lower under irreversibility than without, and so by Lemma A.5.1, funding target increases, making $\widehat{G}_0$ even lower.

In this context, productivity becomes crucial for the incentives for early financing relative to delayed financing as $h_t \to 0^+$, as the following result suggests.

Lemma A.5.2. *$\widehat{G}_0 = -\theta\varphi\overline{h} < 0$ for $A \leq \rho+\delta$. $\frac{d\widehat{G}_0}{dA} \geq 0$ if and only if $\frac{d}{dA}\big(V(\overline{h}) - \overline{h}\big) \geq \frac{1}{\rho+\delta}$.*

Proof. The second claim is immediate from

$$\frac{d}{dA}\frac{\widehat{G}_0}{\theta} = i(\overline{h})\left(\frac{1 - \varphi\frac{d}{dA}\overline{h}}{\rho+\delta - i(\overline{h})}\right) - \varphi\frac{d}{dA}\overline{h} \geq 0 \iff \frac{i(\overline{h})}{\rho+\delta} \geq \varphi\frac{d}{dA}\overline{h}$$

$$\iff \frac{d}{dA}\big(V(\overline{h}) - \overline{h}\big) = \frac{1 - \varphi\frac{d}{dA}\overline{h}}{\rho+\delta - i(\overline{h})} \geq \frac{1 - \frac{i(\overline{h})}{\rho+\delta}}{\rho+\delta - i(\overline{h})} = \frac{1}{\rho+\delta}.$$

[44] To have a well-defined value of V^*, assume that $A \leq (\rho+\delta)\left(\frac{(\rho+\delta)\psi}{2} + 1\right)$.

[45] Given the large gap between $A-\rho-\delta$ and $\frac{1}{2\psi}$, the same conclusion holds as long as divestment at funding target $i(\overline{h}) < 0$ is not severe – that is, $V(\overline{h}) - \overline{h}$ is not too below unity.

The first claim is also immediate since $A \leq \rho+\delta$ implies that investment is not optimal even on a first-best basis, such that $i(\overline{h}) = 0$. $\square$

Given $\underline{A} \equiv \inf\{A > 0 \mid \widehat{G}_0 \geq 0\}$, the first claim above implies that $\frac{d\widehat{G}_0}{dA}\big|_{\underline{A}} \geq 0$ such that

$$\frac{d}{dA}\big(V(\overline{h}) - \overline{h}\big)\Big|_{\underline{A}} \geq \frac{1}{\rho+\delta}.$$

While the proof remains elusive, it is plausible – and consistent with numerical simulations – that $\frac{d}{dA}\big(V(\overline{h}) - \overline{h}\big)\big|_{A} > \frac{1}{\rho+\delta}$ for any $A > \underline{A}$, such that $\widehat{G}_0$ is strictly increasing in $A > \underline{A}$. This is because if A increases marginally, value should in general rise by a factor between $\frac{1}{\rho+\delta}$ and $\frac{1}{\rho+\delta-i(\overline{h})}$, since investment ranges between $[0, i(\overline{h})]$.[46] In fact, consistent with the intuition given that $i(\overline{h})$ is increasing in A, numerical simulation suggests that $V(\overline{h}) - \overline{h}$ is also convex in A given $i(\overline{h}) > 0$; basically, greater investment makes marginal increase in capital productivity more valuable.

[46] In principle, expected future dilution dampens this factor, as shown by the term $-\varphi\frac{d\overline{h}}{dA}$ in

$$\frac{d}{dA}\big(V(\overline{h}) - \overline{h}\big) = \frac{1-\varphi\frac{d\overline{h}}{dA}}{\rho+\delta-i(\overline{h})}.$$

But since net value is being evaluated at the funding target $\overline{h}$, the magnitude of this effect should be relatively small.

Supplemental Appendix A Other Discussions

SA.1 Robustness to within-period bargaining

Here, I discuss how θ can accommodate various static bargaining environments. I illustrate it using the stylized setup in Section 3, but the implications extend to the main model. Let c_t continuation value and v_t^o the firm's reservation value on date t.

Multiple financiers on each date. Suppose that $n \in \mathbb{N}$ financiers visit the firm on each date. I assume that the firm bargains with one financier at a time. If bargaining were to fail, the firm can bargain with another financier on the same date. Each financier commits to refusing to bargain with the firm conditional on his previous bargaining failure with it. Let v_t^m firm value from bargaining with the m^{th} financier on date t, $m \leq n$. Then,

$$
\begin{aligned}
v_t^n &= \theta c_t + (1-\theta) v_t^o, \\
v_t^{n-1} &= \theta c_t + (1-\theta) v_t^n = \big(1-(1-\theta)^2\big) c_t + (1-\theta)^2 v_t^o, \\
v_t^{n-2} &= \theta c_t + (1-\theta) v_t^{n-1} = \big(1-(1-\theta)^3\big) c_t + (1-\theta)^3 v_t^o \\
&\dots \\
v_t^1 &= \theta c_t + (1-\theta) v_t^2 = \big(1-(1-\theta)^n\big) c_t + (1-\theta)^n v_t^o.
\end{aligned}
$$

Therefore, it is equivalent to $\theta_n \equiv \big(1-(1-\theta)^n\big) \in [\theta, 1)$; expectedly, $\theta_n \to 1$ as $n \to \infty$.

Nesting Bertrand competition. Let $n = 2$ above, and suppose that each financier can fully commit to refusing to bargain with the firm consecutively, and otherwise can commit to refusing to bargain with an i.i.d. probability $\chi \in [0,1]$ in case the firm has failed in bargaining beforehand. Once the firm is denied bargaining by either financier, no more bargaining is allowed on that date. Because bargaining does not fail on the path of play in any subgame, the value of each financier's outside options when he is bargaining with the firm is zero. Letting v_t^B firm's value from bargaining,

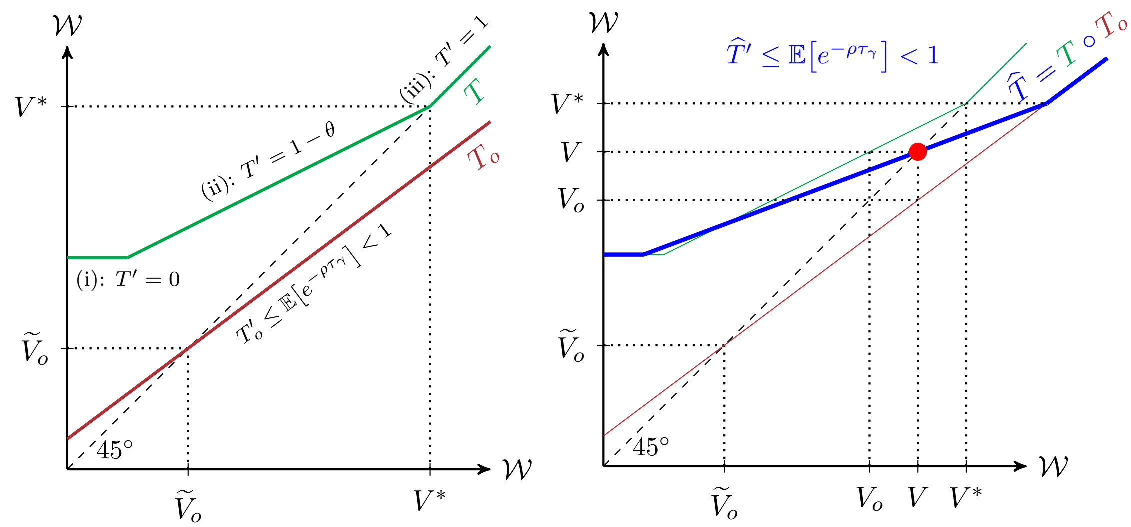
$$
\begin{aligned}
v_t^B &= \theta c_t + (1-\theta)\Big(\chi v_t^o + (1-\chi) v_t^B\Big) \\
\implies v_t^B &= \frac{\theta}{1-(1-\theta)(1-\chi)} c_t + \left(1 - \frac{\theta}{1-(1-\theta)(1-\chi)}\right) v_t^o.
\end{aligned}
$$

Therefore, this is equivalent to $\theta_\chi \equiv \frac{\theta}{1-(1-\theta)(1-\chi)} \in [\theta, 1]$. Note that $\theta_\chi = 1$ if and only if $\chi = 0$, that is, financiers cannot commit at all except upon immediate bargaining failure. This essentially describes Bertrand competition. In sum, when the firm must bargain with each financier bilaterally, perfect competition à la Bertrand between two financiers obtains if and only if financiers lack commitment technology altogether $\chi = 0$.

SA.2 Contraction from two-step recursion: illustration

Take the space $\mathcal{W}$ and the three self-maps $\widehat{T}, T, T_o : \mathcal{W} \to \mathcal{W}$ as defined in Appendix A.1. Here, I illustrate the proof that $\widehat{T}$ is a contraction, such that contraction mapping theorem can be applied. Let $V^* \in \mathcal{W}$ be first-best value function, that is, without financing frictions and $\widetilde{V}_o \in \mathcal{W}$ value function under permanent autarky, that is, $\gamma = 0$.

By way of analogy, visualize the infinite-dimensional $\mathcal{W}$ as a simple $\mathbb{R}_+$. A continuous, nondecreasing, and piece-wise differentiable self-map on $\mathbb{R}_+$ is a contraction if and only if its derivative, wherever defined, is below ζ for some $\zeta \in [0, 1)$.



(a) Before composition T, T_o (b) Composition $\widehat{T}$ and unique fixed point

Figure SA.1: Two-step recursion as a contraction map

Figure SA.1a describes T and T_o. First consider T. If insiders' outside option is preferred to the first-best $W \geq V^*$, they are going to simply take it – the embedded choice (iii) in the definition of T (see Appendix A.1). Therefore, its slope above V^* is unity. If $W < V^*$, then insiders do not take the option, but this 'outside' option is still beneficial to the extent that insiders bargain with financiers. Since better outside option reduces financiers' rents by a fraction $1 - \theta$ subject to bargaining taking place, the slope of T below V^* is (ii) $1 - \theta$ given bargaining and (i) zero given no bargaining. As shown, T is not a contraction: any point above V^* is one of its fixed points.

Next, consider T_o: insiders are on autarky until the stopping time τ_γ when the business gives W as a terminal payoff. A higher terminal payoff benefits insiders but only up to a discounting due to time delay $\mathbb{E}\big[e^{-\rho\tau_\gamma}\big] \in [0, 1)$, which thus bounds the slope of T_o. Therefore, T_o is a contraction and has a unique fixed point.

But T_o, in itself, is a particularly *un*interesting contraction. Regardless of $\gamma \in [0, \infty)$, its fixed point is always $\widetilde{V}_o$. If at τ_γ insiders receive their current value as a terminal payoff, they are exactly compensated for the termination anyway. Accordingly, the graph of T_o rotates as γ changes but centered on the invariant fixed point $(\widetilde{V}_o, \widetilde{V}_o)$.

Even though T_o by itself is a trivial contraction, its time discounting due to $\gamma < \infty$, as Figure SA.1b illustrates, still makes the concatenated map $\widehat{T} = T \circ T_o$ contractionary, because the slope of T never exceeds unity. Hence, there exists a unique – and interesting – fixed point $V = \widehat{T}(V)$ by contraction mapping theorem.

SA.3 Costs and benefits of financial slack

Given a general cash flow profile from Section 4.1, posit the equilibrium $(\underline{h}, \overline{h})$ along with the implied value functions (V, V_o). First, define dilution ratio as

$$D(h) \equiv (1-\theta)\left(1 - \frac{V_o(h) - h}{V(\overline{h}) - \overline{h}}\right),$$

i.e., financiers' rent $(1-\theta)\big(V(\overline{h}) - (\overline{h} - h) - V_o(h)\big)$ over net firm value post financing $V(\overline{h}) - \overline{h}$.[47] It is decreasing in h, and satisfies $1-\theta = D(0) > D(\overline{h}) > 0$. Let $\underline{D} \equiv D(\underline{h})$.

To analyze lifetime firm value without a recursive HJB formulation, a couple of stochastic processes must be specified. Define a counting process $\{n_t\}_{t\geq 0}$ by $n_0 = 0$ and $dn_t = \mathbb{1}\left(\lim_{s\to t^-} h_s = \underline{h}\right)$. The process n_t tracks how many times financing, and hence dilution $\underline{D}$, has occurred over the time interval $(0, t]$. Also define $\{\tau_m\}_{m\in\mathbb{N}}$ as

$$\tau_m \equiv \inf\{t \geq 0 \mid n_t = m\},$$

which is the associated increasing sequence of stopping times for m^{th} financing; that is, the first financing occurs at $t = \tau_1 > 0$, and so forth. Lastly, let τ be the stopping time for terminal success arriving at a Poisson rate λ with payoff Π. Note that n_τ counts the entire lifetime financing rounds; in particular, $\lambda > (=)\ 0$ if and only if $n_\tau \overset{a.s.}{<} (\overset{a.s.}{=})\ \infty$.

Firm value $V(h)$ can be decomposed as follows. Think of it equivalently as $h + \big(V(h) - h\big)$ where insiders, counterfactually, first receive h as one-time dividends and, going forward, recognize both cash flow $\mu\ dt + \sigma\ dB_t$ (including terminal Π at λ) and carry cost flow $-\varphi h_t\ dt$ as immediate utility flow scaled by their undiluted ownership $(1-\underline{D})^{n_t}$. Given $h_0 = \overline{h}$,

$$\begin{aligned} V(\overline{h}) - \overline{h} &= \mathbb{E}_0\left[\int_0^\tau e^{-\rho t}(1-\underline{D})^{n_t}\Big(\big(\mu - \varphi h_t\big)\ dt + \sigma\ dB_t\Big) + e^{-\rho\tau}(1-\underline{D})^{n_\tau}\Pi\right] \\ &= \mathbb{E}_0\left[\int_0^\tau e^{-\rho t}\Big(\mu - \varphi h_t\Big)\ dt + e^{-\rho\tau}\Pi\right] - \mathbb{E}_0\left[\sum_{m=1}^{n_\tau} e^{-\rho\tau_m}\underline{D}\Big(V(\overline{h}) - \overline{h}\Big)\right]. \end{aligned}$$

[47] D and ownership retention x in Section 4.2 satisfy $D(h)\big(V(\overline{h}) - \overline{h}\big) = \big(1 - x(h)\big)V(\overline{h}) - (\overline{h} - h)$.

Inside the second expectation on the last line, both $\underline{D}$ and $V(\overline{h}) - \overline{h}$ are constant in equilibrium for all $\{\tau_m\}_{m\in\mathbb{N}}$ due to the time-invariant financing threshold $\underline{h}$ and funding target $\overline{h}$, and thus can be brought outside the unconditional expectation. Rearranging yields

$$V(\overline{h}) - \overline{h} = \frac{\text{NPV} - \mathcal{C}}{1 + \mathcal{D}}, \tag{SA.1}$$

where

$$\text{NPV} \equiv \mathbb{E}_0\left[\int_0^\tau e^{-\rho t}\mu\ dt + e^{-\rho\tau}\Pi\right] = \frac{\mu + \lambda\Pi}{\rho + \lambda}, \quad \mathcal{C} \equiv \mathbb{E}_0\left[\int_0^\tau e^{-\rho t}\varphi h_t\ dt\right] \equiv \underline{\mathcal{C}} + \mathcal{C}_\Delta$$

with $\underline{\mathcal{C}} \equiv \frac{\varphi}{\rho+\lambda}\underline{h}$ and $\mathcal{C}_\Delta \equiv \varphi\mathbb{E}_0\left[\int_0^\tau e^{-\rho t}(h_t - \underline{h})\ dt\right]$, and

$$\mathcal{D} \equiv \underline{D}\mathbb{E}_0\left[\sum_{m=1}^{n_\tau} e^{-\rho\tau_m}\right].$$

As Equation (SA.1) shows, firm value in net $V(h) - h$, even at the target funding capacity, is lower than the frictionless net present value of the business due to the carry cost of financial slack $\mathcal{C}$ and dilution $\mathcal{D}$. Insiders choose a financing strategy $(\underline{h}, \Delta h)$ to maximize net firm value, balancing the reduction of dilution $\mathcal{D}$ against the carry cost $\mathcal{C}$. On one hand, both funding reserve $\underline{h}$ and financing amount $\Delta h = \overline{h} - \underline{h}$ lower dilution $\mathcal{D}$, by reducing its size $\underline{D}$ and frequency $\mathbb{E}_0\left[\sum_{m=1}^{n_\tau} e^{-\rho\tau_m}\right]$, respectively. On the other hand, financial slack involves carry costs $\mathcal{C} = \underline{\mathcal{C}} + \mathcal{C}_\Delta$, with $\underline{\mathcal{C}}$ from the funding reserve $\underline{h}$ and $\mathcal{C}_\Delta$ from the financing amount Δh.

Financing is always optimally lumpy $\Delta h > 0$ because zero financing amount $\Delta h \to 0$ fails to drive the size of dilution $\underline{D}$ down to zero, as Proposition 1 shows, and yet blows its frequency $\mathbb{E}_0\left[\sum_{m=1}^{n_\tau} e^{-\rho\tau_m}\right]$ up to infinity. In contrast, financing is not always optimally early, as Proposition 2 shows. The funding reserve $\underline{h}$, viewed as insiders' one-dimensional Markov strategy, involves a greater marginal carry cost $\frac{\varphi}{\rho+\lambda}$ than financing amount Δh, because $\varphi\underline{h}\ dt$ is a fixed flow cost whereas $\varphi(h_t - \underline{h})\ dt < \varphi\Delta h\ dt$ almost always. As such, insiders do not employ a funding reserve if it fails to sufficiently reduce the size of dilution $\underline{D}$.

SA.4 Financial slack under investment opportunities

For the present stylized exercise, I slightly modify the cash flow profile in Section 4.1.

Cash flow and investment opportunities. The business has 'normalized' cash inflow $\pi\ dt + \sigma\ dB_t$, with $\pi,\ \sigma > 0$. Opportunities to scale up cash inflow to $\eta\big(\pi\ dt + \sigma\ dB_t\big)$, $\eta > 1$, arrive at a Poisson rate $\lambda > 0$, but it requires a normalized upfront investment *expense* of $\xi > 0$.[48] Firm value $W(k,h)$ given $k \in \mathbb{N} \cup \{0\}$ rounds of past investment and funds h is homogeneous so that $W(k,h) = \eta^k W\left(1, \frac{h}{\eta^k}\right) \equiv \eta^k V\left(\frac{h}{\eta^k}\right)$.

To ensure that investment is first-best, I assume the following.

Assumption SA.4.1. *$\pi > \lambda\xi$ and $\frac{(\eta-1)\pi}{\rho} \geq \xi$.*

Investment choice. Upon receiving an opportunity to invest, a firm with funds h can (i) fund the investment internally, with value $\eta V\left(\frac{h-\xi}{\eta}\right)$, (ii) forgo the investment, with value $V(h)$, or (iii) finance the investment externally, with value given by bargaining as

$$\widehat{V}_o(h) + \theta\Big(\eta V(\overline{h}) - \big(\xi + \eta\overline{h} - h\big) - \widehat{V}_o(h)\Big),$$

where the firm' reservation value is $\widehat{V}_o(h) \equiv \max\left\{V_o(h), \eta V_o\left(\frac{h-\xi}{\eta}\right)\right\}$:

- When $\widehat{V}_o(h) = V_o(h)$, the firm optimally forgoes investment if financing fails.
- When $\widehat{V}_o(h) = \eta V_o\left(\frac{h-\xi}{\eta}\right)$, the firm optimally invests even if financing fails. I term this 'financing the investment with (credible) commitment to execution.'

The term 'commitment' is simply a shorthand for when, given that choice (iii) is optimal, the optimality of investment itself does not depend on successful financing.[49]

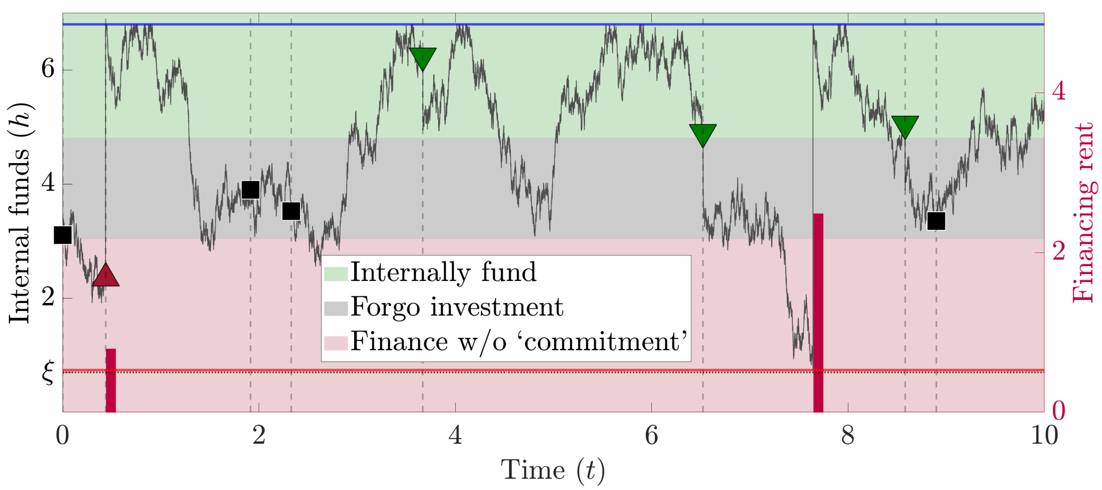


Figure SA.2: Optimal investment policy under dilution from financing

The black curve is a simulated history of normalized internal funds along the left axis. The horizontal lines in blue and red are funding target $\overline{h}$ and financing threshold $\underline{h}$, respectively. Investment opportunities arrive at the vertical dashed lines, each requiring an upfront normalized expense ξ whose size is indicated on the left axis. The markers in each shaded region indicate the choice made upon each opportunity in accordance with investment policy described in the legend. The magenta bar is normalized financing rent: one at investment financing ($t \approx 0.44$), and one at non-investment financing ($t \approx 7.64$). Access to alternative financing is $\gamma = 0.3$.

[48]In the illustration part from Section 6.2 at Figure 9, ξ was fund proceeds from divestment.

[49]By subgame perfection, this is equivalent to the firm first funding the investment internally – per choice (i) – and then promptly raising financing – per Section 4.2.

Financing access. I study how access to alternative financing γ affects financial slack and investment, given $(\rho, r) = (0.07, 0)$, $\theta = 0.5$, $(\pi, \sigma) = (1, 2)$ and $(\lambda, \xi) = (0.5, 0.7)$.

Let $\gamma = 0.3$. As Figure SA.2 illustrates, firms may sometimes forgo – as shown in black square markers – investment opportunities that increase frictionless net present value. For these firms, financing involves large rent extraction as their access to alternative financing is not so good. Therefore, they optimally forgo investment when the lumpy expense would greatly dilute firm value ex-ante by making the sizable rent extraction by financiers much likelier to be triggered soon. When rent extraction has already become highly likely soon due to low funds, however, firm value is already diluted enough ex-ante so that it is optimal to increase diluted firm value by bargaining with financiers over financing the investment.

Figure SA.3 illustrates optimal financing and investment policies as the strength of financing access varies $\gamma \in [0, 52]$. The subplots describe $\gamma \in \{0, 26\}$. In Figure SA.3a with $\gamma = 0$, firms face substantial dilution from financing because they cannot access alternative financiers. As a result, they delay financing as much as possible and often forgo investment in order to avoid large dilution from financing.

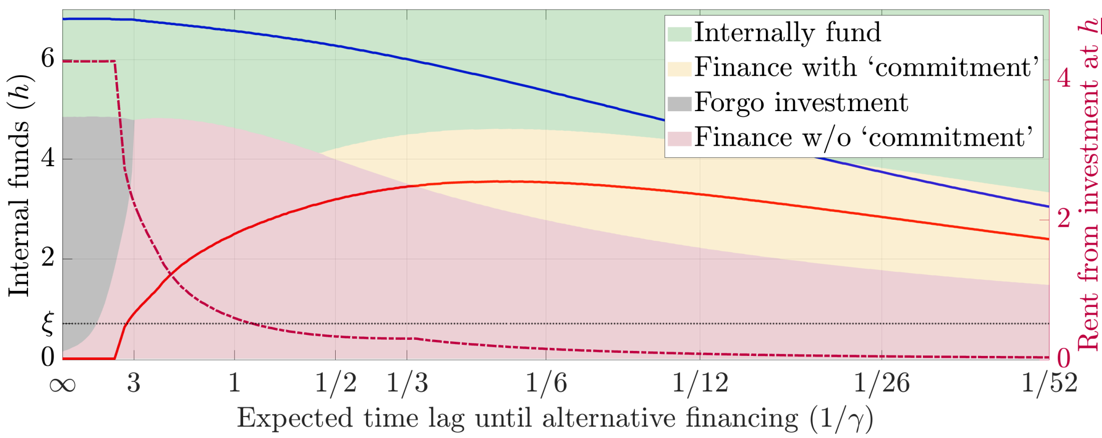
In contrast, Figure SA.3b describes a firm that can expect to find alternative financiers very quickly $\gamma = 26$. It finances in small lumps $\Delta h \approx 0.91$ and thus extremely frequently – 53 times over $t \in [0, 10]$ in the present simulation – because of the extremely negligible size of dilution around 0.004 at the financing threshold. Nevertheless, it maintains a sizable funding reserve $\underline{h} \approx 2.84$. As discussed in Section 5, this financial slack is what reduces dilution to such a negligible size.

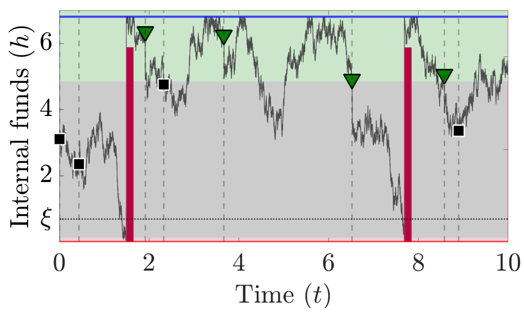
'Commitment' to investment execution and the size of dilution. Moreover, firms with such good access to financing $\gamma = 26$ always finance lumpy investments despite keeping funding reserves $\underline{h} \approx 2.84$ in large excess of the expense $\xi = 0.7$. For any $h_t \in [\underline{h}, \overline{h}]$, these firms are willing to invest even if financing fails: even at $h_t = \underline{h}$,

$$\eta V_o \left(\frac{\underline{h} - \xi}{\eta} \right) \approx 23.228 > 22.102 \approx V_o(\underline{h}). \tag{SA.2}$$

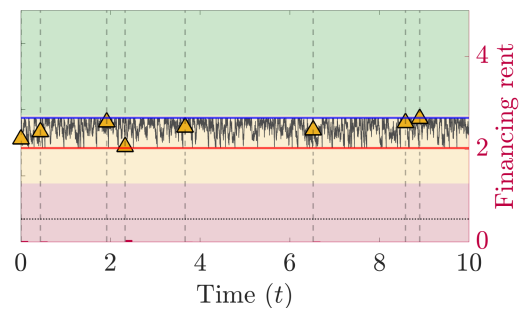
This is because both γ and $\underline{h}$ are high. Because $\underline{h} \approx 2.84$ is high, firms can fund the investment internally and still have a substantial amount of normalized remaining funds $\frac{\underline{h}-\xi}{\eta} \approx 1.95$. Because γ is high, firms' normalized reservation value falls only slightly as a result $V_o \left(\frac{\underline{h}-\xi}{\eta} \right) \approx 21.116 < 22.102 \approx V_o(\underline{h})$. Investment thus raises the actual reservation value, as Inequality (SA.2) shows.

This is how such firms incur negligible dilution even when financing the investment. Even if financing were to fail, investment would still be optimally undertaken. Because investment does not depend on financing, returns on investment do not constitute a surplus from financing. Dilution thus becomes quite negligible (approximately 0.054),





(a) Poor access to financing $\gamma = 0$



(b) Good access to financing $\gamma = 26$

Figure SA.3: Financing access and investment policy

In the main plot with comparative statics in γ, colored areas show optimal investment policy given funds h per the legend. The semi-dashed line in magenta is the size of dilution from financing the investment given the opportunity arrives when $h_t = \underline{h}$. The dilution graph kinks downward when 'commitment' starts to hold – i.e., where the red curve enters the yellow region. For explanation of the subplots, see Figure SA.2.

because this credible 'commitment' reduces it by $(1-\theta)\left(\eta V_o\left(\frac{\underline{h}-\xi}{\eta}\right) - V_o(h)\right) \approx 0.563.$

Supplemental Appendix B Algorithms

Here, I explain the numerical algorithm. I first give a brief sketch of the algorithm for the main model in Sections 4 and 5, and then provide details in the context of investment choice in Section 6. As for the investment algorithm, Supplemental Appendix SB.1 sets up the main algorithm for the case without irreversibility and fluctuations in parameters. I then introduce Markov chains, both discrete and continuous, in Supplemental Appendix SB.2. Last, Supplemental Appendix SB.3 describes how to solve the model under investment irreversibility.

Sketch of the algorithm. The key idea of the algorithm is that the following differential equations must be *globally* satisfied: given sufficiently high $H > 0$,

$$
\begin{aligned}
&(\rho+\lambda)V(h) - (\mu + rh)V'(h) - \frac{1}{2}\sigma^2 V''(h) = \mathrm{NL}(V, V_o, h) \\
&\qquad \equiv \max\Big\{(\rho+\lambda)V(h) - (\mu+rh)V'(h), \lambda\big(\Pi + h\big), \\
&\qquad\qquad (\rho+\lambda)\Big(\theta\big(V(H) - H + h\big)(1-\theta)V_o(h)\Big) - (\mu+rh)V'(h) - \frac{1}{2}\sigma^2 V''(h)\Big\}, \\
&(\rho+\lambda+\gamma)V_o(h) - (\mu+rh)V_o'(h) - \frac{1}{2}\sigma^2 V_o''(h) = \mathrm{NL}_o(V, V_o, h) \\
&\qquad \equiv \max\Big\{(\rho+\lambda+\gamma)V_o(h) - (\mu+rh)V_o'(h), \lambda\big(\Pi+h\big) + \gamma V(h)\Big\}
\end{aligned}
$$

for all $h \in (0, H)$, along with the boundary conditions

$$
V(0) = \theta\big(V(H) - H\big), \quad V'(H) = 1, \quad V_o(0) = 0, \quad V_o'(H) = 1.
$$

The first element in NL is the maximum for $h \in (\overline{h}, H)$, the second for $h \in (0, \overline{h})\backslash B$, and the third for $h \in B$. Similarly, the first element in NL_o is the maximum for $h \in (\overline{h}_o, H)$ and the second for $h \in (0, \overline{h}_o)$. 'NL' stands for 'nonlinear in (V, V_o).'

Given initial guess for (V^{old}, V_o^{old}), compute NL and NL_o as vectors with respect h, use linear solver to get (V^{new}, V_o^{new}) solving the above, and iterate until convergence.

One benefit of this approach is that it does not impose Lemma 2. This is useful in Section 6 because, due to intractability introduced by investment choice, it is elusive to analytically establish monotone financing strategy in the same fashion. Numerical results displayed in Section 6 do have monotone financing strategy when financing is early, not because it is imposed but because it is numerically predicted.

SB.1 Main algorithm for investment extension

Formulation. Start by setting some $H > 0$. It should be higher than $\overline{h}^o$, the funding target under exclusion, which is higher than $\overline{h}$ but typically by a slight margin. V on

$[0, H]$ satisfies:

$$h \geq \overline{h} \implies 0 = V_{hh}(h) \quad (\because V_h = 1 \text{ on } [\overline{h}, \infty)) \tag{SB.1}$$

$$\begin{aligned} h \in [\underline{h}, \overline{h}] \implies \rho V(h) - rhV_h(h) &= \max_i \big(A - i - \Psi(i)\big)V_h + \big(i - \delta\big)\big(V - hV_h\big) + \frac{1}{2}\sigma^2 V_{hh} \\ &= \left(A + \frac{1}{2\psi} + \left(\delta + \frac{1}{\psi}\right)h\right)V_h - \left(\delta + \frac{1}{\psi}\right)V + \frac{1}{2\psi}\frac{(V - hV_h)^2}{V_h} + \frac{1}{2}\sigma^2 V_{hh} \end{aligned} \tag{SB.2}$$

$$h \leq \underline{h} \implies V(h) = \theta\Big(V(H) - H + h\Big) + (1 - \theta)V^o(h). \tag{SB.3}$$

Note that (SB.1) implies $V(H) - H = V(\overline{h}) - \overline{h}$, which is being substituted into (SB.3). Next, V^o on $[0, H]$ satisfies:

$$h \geq \overline{h}^o \implies 0 = V^o_{hh}(h) \quad (\because V^o_h = 1 \text{ on } [\overline{h}^o, \infty)) \tag{SB.4}$$

$$\begin{aligned} h \in [0, \overline{h}^o] \implies & \rho V^o(h) - rhV^o_h(h) \\ &= \left(A + \frac{1}{2\psi} + \left(\delta + \frac{1}{\psi}\right)h\right)V^o_h - \left(\delta + \frac{1}{\psi}\right)V^o + \frac{1}{2\psi}\frac{(V^o - hV^o_h)^2}{V^o_h} + \frac{1}{2}\sigma^2 V^o_{hh} \\ &\quad + \gamma\big(V(h) - V^o(h)\big). \end{aligned} \tag{SB.5}$$

For ease of notation, define

$$\alpha \equiv \rho + \delta + \frac{1}{\psi}, \quad \beta(h) \equiv A + \frac{1}{2\psi} + \left(r + \delta + \frac{1}{\psi}\right)h, \quad \xi(v, v_h, h) \equiv \frac{1}{2\psi}\frac{(v - hv_h)^2}{v_h}.$$

The five piecewise equalities above – (SB.1) through (SB.5) – switch to strict inequalities when evaluated outside the respective intervals, with left-hand sides being higher. Therefore, these can be summarized as follows: for $h \in [0, H]$,

$$\begin{aligned} &\alpha V(h) - \beta(h)V_h(h) - \frac{1}{2}\sigma^2 V_{hh}(h) = \mathrm{NL}(V, V^o, h) \\ &\equiv \max\Bigg\{\alpha V(h) - \beta(h)V_h(h), \xi\big(V(h), V_h(h), h\big), \\ &\qquad \alpha\Big(\theta\big(V(H) - H + h\big) + (1 - \theta)V^o(h)\Big) - \beta(h)V_h(h) - \frac{1}{2}\sigma^2 V_{hh}(h)\Bigg\}, \end{aligned} \tag{SB.6}$$

$$\begin{aligned} &(\alpha + \gamma)V^o(h) - \beta(h)V^o_h(h) - \frac{1}{2}\sigma^2 V^o_{hh}(h) = \mathrm{NL}^o(V, V^o, h) \\ &\qquad \equiv \max\Big\{(\alpha + \gamma)V^o(h) - \beta(h)V^o_h(h), \xi\big(V^o(h), V^o_h(h), h\big) + \gamma V(h)\Big\}. \end{aligned} \tag{SB.7}$$

Both NL and NL^o capture the nonlinear components of the pair of differential equations. In the expression for NL, the first element gives the maximum on $[\overline{h}, H]$, the second on $[\underline{h}, \overline{h}]$ and the last on $[0, \underline{h}]$, and similarly in NL^o given $\underline{h}^o = 0$.[50] Last, the boundary conditions are:

$$V(0) = \theta\Big(V(H) - H\Big),\ V_h(H) = 1 \tag{SB.8}$$

$$V^o(0) = 0,\ V_h^o(H) = 1. \tag{SB.9}$$

Discretization and linearization. Let us discretize the fund space $[0, H]$ into N_h evenly spaced grids, and let $\Delta H \equiv \frac{H}{N_h - 1}$ the grid size. For now, let $i \in \{1, 2, \ldots, N_h\}$ index $[0, H]$ increasingly, and denote $h \in [0, H]^{N_h}$ as the column vector discretizing $[0, H]$, such that $h(0) = 0,\ h(N_h) = H$. Posit $V_0, V_0^o \in \mathbb{R}^{N_h}$ as column vectors representing conjectured approximate value functions under inclusion and exclusion, respectively. Let $W_0 \in \mathbb{R}^{2N_h}$ with

$$W_0 \equiv \begin{pmatrix} V_0 \\ V_0^o \end{pmatrix},$$

represent the stacked value functions. $i \in \{N_h + 1, \ldots, 2N_h\}$ represents funds under exclusion.

The core of the algorithm is to summarize the left-hand sides of the combined HJB equations (SB.6), (SB.7) as well as the boundary conditions (SB.8), (SB.9) into a single $2N_h \times 2N_h$ sparse matrix $M(W)$, which depends on the true stacked value function W, such that $M(W) \cdot W = \mathbf{NL}(W)$, where $\mathbf{NL}$ is essentially a stack of NL and NL^o but with some additional adjustments at the respective end rows, to be specified soon. M should depend on W exclusively due to the upwinding method described shortly. I will start with some initial W_0 and use a linear solver to obtain W_1 such that

$$\frac{W_1 - W_0}{\Delta t} + M(W_0) \cdot W_1 = \mathbf{NL}(W_0).$$

Above, the first term on the left-hand side is the pseudo-time derivative to enable efficient convergence, with $\Delta t \in \mathbb{R}_{++}$. Until W_1 is sufficiently close to W_0, update W_0 and repeat.

In principle, $M(\cdot)$ is a mapping from $\mathbb{R}^{2N_h}$ to $\mathbb{R}^{2N_h \times 2N_h}$, but the sparsity substantially reduces its rank. This mapping should capture everything linear in both the combined HJB equations, including the linear differential terms, and the boundary conditions.

[50] The first elements of NL and NL^o make use of the fact that $V_{hh}, V_{hh}^o \leq 0$ with the equality if and only if $h \geq \overline{h},\ h \geq \overline{h}^o$, respectively. The second elements are the standard HJB equations. The third element in NL derives from $V \geq \theta\big(V(H) - H + h\big) + (1 - \theta)V^o$, with the equality if and only if $h \leq \underline{h}$.

Upwinding the derivatives. I will approximate the combined HJB equations (SB.6), (SB.7) for the interior rows of h only, i.e., $i = 2, 3, \ldots, N_h - 1$. The endpoints $i \in \{1, N_h\}$ will be reserved for the boundary conditions (SB.8), (SB.9). For each of the interior rows $i \in \{2, 3, \ldots, N_h - 1\}$, I approximate the first and second derivatives of a given approximated function V as follows: define

$$\Delta_h^f \equiv \left(0, -\frac{1}{\Delta H}, \frac{1}{\Delta H}\right), \quad \Delta_h^b \equiv \left(-\frac{1}{\Delta H}, \frac{1}{\Delta H}, 0\right),$$
$$\Delta_h^c \equiv \left(-\frac{1}{2\Delta H}, 0, \frac{1}{2\Delta H}\right), \; \Delta_h^2 \equiv \left(\frac{1}{\Delta H^2}, -\frac{2}{\Delta H^2}, \frac{1}{\Delta H^2}\right).$$

Then, letting $\widehat{V}(i) \equiv \Big(V(i-1), V(i), V(i+1)\Big)'$, $\Delta_h^f \cdot \widehat{V}(i)$ denotes the forward first difference, $\Delta_h^b \cdot \widehat{V}(i)$ the backward first difference, $\Delta_h^c \cdot \widehat{V}(i)$ the centered first difference, and $\Delta_h^2 \cdot \widehat{V}(i)$ the second difference, all in h.

I follow the standard numerical method of 'upwinding' where the forward/backward difference is used in approximating the first derivative with a positive/negative drift. Determining the sign of the drift in cash flow, however, is somewhat tricky since it depends on which region – $[0, \underline{h}], (\underline{h}, \overline{h}]$ or $(\overline{h}, H]$ under inclusion and $[0, \overline{h}^o]$ or $(\overline{h}^o, H]$ under exclusion – the current internal funds level belongs to. Determining the region is equivalent to determining which element is the maximum in NL and NL^o in Equations (SB.6) and (SB.7), respectievly, whose elements all involve first derivatives.

Therefore, I employ the centered first difference to determine the regions and then use them to implement the upwinding. Given V_0 and V_0^o, define ten row indicator vectors – $f_m, b_m \in \{0,1\}^{N_h}$ for $m = 1, 2, 3$ and $f_m^o, b_m^o \in \{0,1\}^{N_h}$ for $m = 1, 2$ – such that $i \in \{1, N_h\} \implies \forall m,\; f_m(i) = b_m(i) = f_m^o(i) = b_m^o(i) = 0$ and for $i \in \{2, 3, \ldots, N_h - 1\}$, $f_1(i) = f_1^o(i) = 1$,

$$f_2(i) \equiv \mathbb{1}\Bigg(A - \mathrm{E}\Big(V_0(i), \Delta_h^c \cdot \widehat{V}_0(i), h(i)\Big) + \Big(r + \delta - \mathrm{I}\Big(V_0(i), \Delta_h^c \cdot \widehat{V}_0(i), h(i)\Big)\Big)h(i) > 0\Bigg),$$

$$f_2^o(i) \equiv \mathbb{1}\Bigg(A - \mathrm{E}\Big(V_0^o(i), \Delta_h^c \cdot \widehat{V}_0^o(i), h(i)\Big) + \Big(r + \delta - \mathrm{I}\Big(V_0^o(i), \Delta_h^c \cdot \widehat{V}_0^o(i), h(i)\Big)\Big)h(i) > 0\Bigg),$$

$$\begin{aligned} f_3(i) \equiv \mathbb{1}\Bigg(&A - \mathrm{E}\Big(V_0^o(i) - \theta\big(V_0(N_h) - H + h(i)\big), \Delta_h^c \cdot \widehat{V}_0(i) - \theta, h(i)\Big) \\ &+ \Big(r + \delta - \mathrm{I}\Big(V_0^o(i) - \theta\big(V_0(N_h) - H + h(i)\big), \Delta_h^c \cdot \widehat{V}_0(i) - \theta, h(i)\Big)h(i)\Big) > 0\Bigg), \end{aligned}$$

and $b_m(i) = 1 - f_m(i)$, $b_m^o(i) = 1 - f_m^o(i)$ for all m. Above, I and E denote optimized values of gross investment i[51] and total investment expense $i + \Psi(i)$, respectively, given

[51] I and δ enter the drift in cash flow because h is internal funds per capital.

as

$$\mathrm{I}(v, v_h, h) = \frac{1}{\psi}\left(\frac{v}{v_h} - h - 1\right), \quad \mathrm{E}(v, v_h, h) = \frac{1}{2\psi}\left(\left(\frac{v - hv_h}{v_h}\right)^2 - 1\right).$$

As can be inferred from $f_1 = f_1^o = 1$, I use forward differences on $(\overline{h}, H]$ and $(\overline{h}^o, H]$. Let

$$\Delta_{hm}(i) \equiv f_m(i)\Delta_h^f + b_m(i)\Delta_h^b, \quad \Delta_{hmo}(i) \equiv f_m^o(i)\Delta_h^f + b_m^o(i)\Delta_h^b.$$

Let $m^*(i) \in \{1, 2, 3\}$ and $m^{o*}(i) \in \{1, 2\}$ index the maximum, respectively in the sets

$$\Big\{\alpha V_0(i) - \beta(h(i))\Delta_{h1}(i) \cdot \widehat{V}_0(i), \xi\big(V_0(i), \Delta_{h2}(i) \cdot \widehat{V}_0(i), h(i)\big), \alpha\Big(\theta(V_0(N_h) - H + h(i)) + (1-\theta)V_0^o(i)\Big) - \beta(h(i))\Delta_{h3}(i) \cdot \widehat{V}_0(i) - \frac{1}{2}\sigma^2\Delta_h^2 \cdot \widehat{V}_0(i)\Big\}, \tag{SB.10}$$

$$\Big\{(\alpha+\gamma)V_0^o(i) - \beta(h(i))\Delta_{h1o}(i) \cdot \widehat{V}_0^o(i), \xi\big(V_0^o(i), \Delta_{h2o}(i) \cdot \widehat{V}_0^o(i), h(i)\big) + \gamma V_0(i)\Big\}. \tag{SB.11}$$

Last, let $f(i) \equiv f_{m^*(i)}(i)$, $f^o(i) \equiv f^o_{m^{o*}(i)}(i)$, $b(i) \equiv 1 - f(i)$, $b^o(i) \equiv 1 - f^o(i)$, and

$$\Delta_h(i) \equiv f(i)\Delta_h^f + b(i)\Delta_h^b, \quad \Delta_h^o(i) \equiv f^o(i)\Delta_h^f + b^o(i)\Delta_h^b.$$

These three-dimensional row vectors $\Delta_h(i),\ \Delta_h^o(i) \in \mathbb{R}^3$ implement the upwinding for first differences in the construction of $M(\cdot)$, to which I now transition.

Constructing the matrix. As a reminder, $M(W_0)$ is a $2N_h \times 2N_h$ sparse matrix, because W_0 is a stacked vector of V_0, V_0^o. For each of $i = 2, 3, \ldots, N_h - 1$ rows, let

$$M(i, i-1 : i+1 \mid W_0) \equiv (0, \alpha, 0) - \beta(h(i))\Delta_h(i) - \frac{1}{2}\sigma^2\Delta_h^2,$$

where $M(i, i-1 : i+1)$ denotes the i^{th} row from the $(i-1)^{\text{th}}$ column to the $(i+1)^{\text{th}}$ column, to account for the left-hand side of Equation (SB.6). Similarly, for each of $(N_h + i)^{\text{th}}$ rows with $i = 2, \ldots, N_h - 1$, implement the left-hand side of (SB.7) by

$$M(N_h + i, N_h + i - 1 : N_h + i + 1 \mid W_0) \equiv (0, \alpha + \gamma, 0) - \beta(h(i))\Delta_h^o(i) - \frac{1}{2}\sigma^2\Delta_h^2.$$

Next, construct the 'nonlinear' column vector $\mathbf{NL}(W_0) \in \mathbb{R}^{2N_h}$ as follows: for each of $i = 2, 3, \ldots, N_h - 1$ rows, $\mathbf{NL}(i \mid W_0)$ is the maximum in the set (SB.10) and $\mathbf{NL}(N_h + i \mid W_0)$ in the set (SB.11). The rows $i = 1, N_h, N_h + 1, 2N_h$ will be separately specified right below.

Last, the boundary conditions (SB.8) and (SB.9) are implemented as:

$$M(1,1 \mid W_0) \equiv \alpha,\ M(1,N_h \mid W_0) \equiv -\theta\left(\frac{1}{\Delta t}+\alpha\right),\ \mathbf{NL}(1 \mid W_0) \equiv -\theta\left(\frac{1}{\Delta t}+\alpha\right)H,$$

$$M(N_h, N_h-1 \mid W_0) \equiv -\left(\frac{1}{\Delta t}+\alpha\right),\ M(N_h,N_h \mid W_0) \equiv \alpha,$$

$$\mathbf{NL}(N_h \mid W_0) \equiv \left(\frac{1}{\Delta t}+\alpha\right)\Delta H,$$

$$M(N_h+1, N_h+1 \mid W_0) \equiv 0,\ \mathbf{NL}(N_h+1 \mid W_0) \equiv 0,$$

$$M(2N_h, 2N_h-1 \mid W_0) \equiv -\left(\frac{1}{\Delta t}+\alpha+\gamma\right),\ M(2N_h,2N_h \mid W_0) \equiv \alpha+\gamma,$$

$$\mathbf{NL}(2N_h \mid W_0) \equiv \left(\frac{1}{\Delta t}+\alpha+\gamma\right)\Delta H.$$

Any unspecified element of $M(W_0)$ is set to zero, making it highly sparse.

Iteration to solution. Posit H. Start with some initial guess V_0, V_0^o, and stack them into W_0. Obtain W_1 that solves

$$\left(\frac{1}{\Delta t}\mathbb{I}_{2N_h} + M(W_0)\right)\cdot W_1 = \frac{1}{\Delta t}W_0 + \mathbf{NL}(W_0),$$

where $\mathbb{I}_{2N_h}$ is the $2N_h \times 2N_h$ identity matrix, also highly sparse. If W_1 is close enough to W_0, stop; $V \equiv W_1(1:N_h)$ and $V^o \equiv W_1(N_h+1:2N_h)$. Otherwise, update $W_0 \equiv aW_1 + (1-a)W_0$ for some weight $a \in (0,1]$ and repeat.

If H is set too high, the algorithm might converge too slowly and the solution becomes unnecessarily coarse on $[0,\overline{h}]$. On the other hand, if H is too low, then possibly $H < \overline{h}^o$, in which case the algorithm fails. Therefore, I run the algorithm twice, an initializer and a verifier. During initialization, I use an adequate fraction (say 0.2) of the first-best value V^* as the initial H and use a high error tolerance. If convergence fails, I raise the initial H. If it succeeds, I choose a new H to be only slightly higher than $\overline{h}^o$ and run the verifier with a low error tolerance.

SB.2 Fluctuations in parameters

Stacked value. Let N_s denote the number of Markov states. For a continuous Markov chain, N_s is the number of grids in state-space discretization. I use $s \in \{1,2,\ldots,N_s\}$ to index the Markov states. Given $\left(V_0(i_h,s), V_0^o(i_h,s)\right)$ for $(i_h,s) \in \{1,\ldots,N_h\} \times \{1,\ldots,N_s\}$, define

$$W_0 \equiv \left(V_0(:,1)' \quad \ldots \quad V_0(:,N_s)' \quad V_0^o(:,1)' \quad \ldots \quad V_0^o(:,N_s)'\right)' \in \mathbb{R}^{2N_hN_s}$$

as their stacked column vector. $i \in \{1, \dots, 2N_h N_s\}$ now jointly indexes (i_h, s) and inclusion/exclusion. The mapping $M : \mathbb{R}^{2N_h N_s} \to \mathbb{R}^{2N_h N_s \times 2N_h N_s}$ will be defined in a fashion overall identical to Section SB.1 for each of the $N_h \times N_h$ blocks corresponding to $s \in \{1, 2, \dots, N_s\}$ along the main diagonal $M\big((s-1)N_h + 1 : sN_h, (s-1)N_h + 1 : sN_h \mid W_0\big)$. There will be, however, an additional sparse matrix for the Markov chain and some changes to NL, NL^o.

Discrete Markov chain. Consider a Markov chain in Poisson arrival rates of transition given as

$$\begin{pmatrix} -\lambda^1 & \lambda^1_2 & \dots & \lambda^1_{N_s} \\ \lambda^2_1 & -\lambda^2 & \dots & \lambda^2_{N_s} \\ \dots & \dots & \dots & \dots \\ \lambda^{N_s}_1 & \lambda^{N_s}_2 & \dots & -\lambda^{N_s} \end{pmatrix},$$

where $\lambda^s_{s'} \geq 0$ is the Poisson rate of transition at s to $s' \neq s$, and $\lambda^s \equiv \sum_{s' \neq s} \lambda^s_{s'}$.

Modify the combined HJB equations (SB.6) and (SB.7): for $s \in \{1, \dots, N_s\}$,

$$\begin{aligned}
&(\alpha(s) + \lambda^s)V(h,s) - \beta(h,s)V_h(h,s) - \frac{1}{2}\sigma^2 V_{hh}(h,s) - \sum_{s' \neq s} \lambda^s_{s'} V(h,s') = \mathrm{NL}(V, V^o, h, s) \\
&\equiv \max\Big\{(\alpha(s) + \lambda^s)V(h,s) - \beta(h,s)V_h(h,s) - \sum_{s' \neq s} \lambda^s_{s'} V(h,s'), \xi\big(V(h,s), V_h(h,s), h, s\big), \\
&\quad (\alpha(s) + \lambda^s)\Big(\theta(s)\big(V(H,s) - H + h\big) + (1 - \theta(s))V^o(h,s)\Big) - \beta(h,s)V_h(h,s) \\
&\qquad - \frac{1}{2}\sigma^2 V_{hh}(h,s) - \sum_{s' \neq s} \lambda^s_{s'} V(h,s')\Big\},
\end{aligned} \tag{SB.12}$$

$$\begin{aligned}
&(\alpha(s) + \lambda^s + \gamma(s))V^o(h,s) - \beta(h,s)V^o_h(h,s) - \frac{1}{2}\sigma^2 V^o_{hh}(h) - \sum_{s' \neq s} \lambda^s_{s'} V^o(h,s') \\
&= \mathrm{NL}^o(V, V^o, h, s) \equiv \max\Big\{(\alpha(s) + \lambda^s + \gamma(s))V^o(h,s) - \beta(h,s)V^o_h(h,s) \\
&\qquad - \sum_{s' \neq s} \lambda^s_{s'} V^o(h,s'), \xi\big(V^o(h,s), V^o_h(h,s), h, s\big) + \gamma(s)V(h)\Big\}.
\end{aligned} \tag{SB.13}$$

The dependence of $\alpha, \beta, \theta, \gamma, \xi$ on s captures the fluctuations in parameters.

Define a Markov chain matrix for the entire (i_h, s) space by

$$\Lambda \equiv \begin{pmatrix} -\lambda^1 \widetilde{\mathbb{I}}_{N_h} & \lambda^1_2 \widetilde{\mathbb{I}}_{N_h} & \dots & \lambda^1_{N_s} \widetilde{\mathbb{I}}_{N_h} \\ \lambda^2_1 \widetilde{\mathbb{I}}_{N_h} & -\lambda^2 \widetilde{\mathbb{I}}_{N_h} & \dots & \lambda^2_{N_s} \widetilde{\mathbb{I}}_{N_h} \\ \dots & \dots & \dots & \dots \\ \lambda^{N_s}_1 \widetilde{\mathbb{I}}_{N_h} & \lambda^{N_s}_2 \widetilde{\mathbb{I}}_{N_h} & \dots & -\lambda^{N_s} \widetilde{\mathbb{I}}_{N_h} \end{pmatrix} \in \mathbb{R}^{N_h N_s \times N_h N_s},$$

where $\widetilde{\mathbb{I}}_{N_h}$ is the $N_h \times N_h$ identity matrix but with the first and last main diagonal elements replaced with zero; the first and the last rows in each block are preserved for the boundary conditions. Proceed to extend Λ to both inclusion and exclusion by defining

$$\mathbf{\Lambda} \equiv \begin{pmatrix} \Lambda & \mathbf{0} \\ \mathbf{0} & \Lambda \end{pmatrix} \in \mathbb{R}^{2N_h N_s \times 2N_h N_s}.$$

The zero off-diagonal blocks indicate that Λ is orthogonal to re-inclusion upon γ.

The construction of each main diagonal block of $M(W_0)$ – i.e., $M\big((s-1)N_h + 1 : sN_h, (s-1)N_h + 1 : sN_h \mid W_0\big)$ – is unchanged, both for the interior rows $i \in \{(s-1)N_h + 2, \ldots, sN_h - 1\}$ that implement the combined HJB (SB.12), (SB.13)[52] and for the boundaries $i \in \{(s-1)N_h + 1, sN_h\}$ that implement the same boundary conditions. $\mathbf{NL}(W_0)$ is adjusted slightly for the interior, following the modified definition of $\mathrm{NL}, \mathrm{NL}^o$ in (SB.12), (SB.13). Once $M(\cdot)$, $\mathbf{NL}(\cdot)$ are constructed, iteratively solve

$$\left(\frac{1}{\Delta t}\mathbb{I}_{2N_h N_s} + M(W_0) - \mathbf{\Lambda}\right) \cdot W_1 = \frac{1}{\Delta t} W_0 + \mathbf{NL}(W_0).$$

Continuous Markov chain. Let s_t follow $ds_t = \mu_s(s_t)\ dt + \sigma_s(s_t)\ dZ_t$. Discretize the state space into N_s grids with size ΔS. Let $s \in \{1, 2, \ldots, N_s\}$ index the state space increasingly. The above law of motion is 'discretized' as a Markov chain:

$$\begin{aligned} \mu_s(s) \geq 0 &\implies \lambda^s_{s-1} = \frac{\sigma_s(s)^2}{2\Delta S^2}, \quad \lambda^s_{s+1} = \frac{\mu_s(s)}{\Delta S} + \frac{\sigma_s(s)^2}{2\Delta S^2}, \\ \mu_s(s) < 0 &\implies \lambda^s_{s-1} = -\frac{\mu_s(s)}{\Delta S} + \frac{\sigma_s(s)^2}{2\Delta S^2}, \quad \lambda^s_{s+1} = \frac{\sigma_s(s)^2}{2\Delta S^2}. \end{aligned}$$

As for the endpoints $s \in \{1, N_s\}$, mean reversion will generally allow upwinding of the first-order terms $\pm\frac{\mu_s(s)}{\Delta S}$. The second-order terms, however, cannot be correctly computed, as they go outside the grid. I therefore use W_0 to compute second as well as third finite differences at $s \in \{2, N_s - 1\}$ and use them to linearly approximate the endpoint second derivatives.[53]

Once the discretized Markov chain is set up, follow the same procedure as above.

SB.3 Investment irreversibility

The algorithm remains mostly the same and is modified only as follows. First, whenever $\frac{1}{\psi}$ appears in the above algorithm, multiply it by $1 + (\phi - 1) \cdot \mathbb{1}(\frac{V_0}{\Delta_h \cdot V_0} - h < 1)$; use V_0^o, Δ_h^o instead of V_0, Δ_h when appropriate. The indicator function tracks whether the firm divests. Second, move all terms involving this modified expression to the inside of $\mathbf{NL}$ instead of M, as investment versus divestment can make the system volatile.

[52] The maximizing indices m^*, m^{o*} are based on the modified NL and NL^o in (SB.12), (SB.13).

[53] At the endpoints, the 'discrete jump' interpretation might not hold: the algorithm still works.

Supplemental Appendix C Additional Analysis

This Appendix provides additional analysis for a particularly tractable case of a startup (see Section 4.1) with zero internal yield $r = 0$. Zero internal yield allows analytic solutions for the HJB equations in Section 4.3, and zero volatility $\sigma = 0$ as specified for a startup eliminates the option effect discussed in Section 5.4, streamlining the algebra.

A key goal of this exercise is to demonstrate that a firm's *future* prospect – that is, Π – increases financial slack.[54]

SC.1 Closed-form solution procedure

1. In all cases, $V'(\overline{h}) = 1$.
2. Solve the model with $\gamma = 0$. By Proposition 2, $\underline{h} = 0$, and $\overline{h}$ is implicitly defined by $V(0) = \theta\big(V(\overline{h}) - \overline{h}\big)$.
3. For $\gamma > 0$, first determine whether $\underline{h} = 0$ or $\underline{h} > 0$. This can be done as follows: (i) posit the value of $\overline{h}$ obtained in Step 2, and (ii) evaluate Inequality (11) for $h = 0$. If it fails, then set $\underline{h} = 0$ and assign to $\overline{h}$ the value obtained in Step 2.
4. If the inequality holds, then it must be that $\underline{h} > 0$.[55] Use the following conditions to implicitly determine $(\underline{h}, \overline{h})$:
   (a) Stationary Recursion: $V(\overline{h}) - V(\underline{h}) = \left(1 + \frac{\varphi}{\gamma}\right) \Delta h$, and
   (b) Smooth Pasting: $V'(\underline{h}) = \theta + (1-\theta)V_o'(\underline{h})$.

SC.2 Analytic equilibrium solution

The business incurs a fixed flow expense $\kappa\, dt$, $\kappa > 0$, until success arrives at Poisson rate $\lambda > 0$ upon which the business terminates with one-time payoff $\Pi > 0$. As discussed, assume $r = 0$. Let us reiterate the first part of Assumption 1 as a reference.

Assumption SC.2.1 (Positive net present value)**.** *$\lambda\Pi > \kappa$.*

Then, V on $(\underline{h}, \overline{h})$ satisfies the following ODE:

$$\rho V(h) = \lambda\big(\Pi + h - V(h)\big) - \kappa V'(h)$$
$$\implies V(h) = -ce^{-\frac{\rho+\lambda}{\kappa}h} + \frac{\lambda}{\rho+\lambda}\left(\Pi + h - \frac{\kappa}{\rho+\lambda}\right),$$

[54] The result is consistent with Section 3 in that the two conditions for financial slack – Inequalities (1) and (2) – are easier to satisfy when terminal payoff v is higher.

[55] This is not necessarily the case if $\sigma > 0$ due to the option effect.

for some $c \in \mathbb{R}$. In addition, since $V'(\overline{h}) = 1$, we have

$$V(h) = \frac{1}{\rho+\lambda}\left(\lambda\Big(\Pi + h\Big) - \frac{\kappa}{\rho+\lambda}\left(\lambda + \rho e^{\frac{\rho+\lambda}{\kappa}(\overline{h}-h)}\right)\right), \tag{SC.1}$$

$$V(\overline{h}) = \frac{1}{\rho+\lambda}\left(\lambda\Big(\Pi + \overline{h}\Big) - \kappa\right). \tag{SC.2}$$

SC.2.1 Baseline: no re-inclusion

First consider $\gamma = 0$. Since $\underline{h} = 0$ by Proposition 2, the equilibrium – just $\overline{h}$ in this case – is implicitly defined by the stationary recursion as follows:

$$V(0) = x(0)V(\overline{h}) = \theta(V(\overline{h}) - \overline{h})$$

$$\iff \theta\rho\big((\rho+\lambda)\overline{h} + \kappa\big) + (1-\theta)\lambda\big((\rho+\lambda)\Pi - \kappa\big) = \rho\kappa \exp\left(\frac{\rho+\lambda}{\kappa}\overline{h}\right). \tag{SC.3}$$

In the first line, $V_o(0) = 0$ is used. Note that the solution to Equation SC.3 is positive if and only if Assumption SC.2.1 holds.

With Equation (SC.3), comparative statics is straightforward.

Proposition SC.2.1 (Comparative statics – Startups without re-inclusion)**.**

$$\frac{\partial \overline{h}}{\partial \theta} = -\frac{1}{\rho}\frac{\lambda\Pi - \kappa - \rho\overline{h}}{\exp\left(\frac{\rho+\lambda}{\kappa}\overline{h}\right) - \theta} < 0, \ \lim_{\theta\to 0}\overline{h} < \frac{\lambda\Pi - \kappa}{\rho}, \ \lim_{\theta\to 1}\overline{h} = 0, \quad \textit{and} \ \lim_{\theta\to 1}\frac{\partial \overline{h}}{\partial \theta} = -\infty, \tag{SC.4}$$

$$\frac{\partial \overline{h}}{\partial \Pi} = \frac{\lambda}{\rho}\frac{1-\theta}{\exp\left(\frac{\rho+\lambda}{\kappa}\overline{h}\right) - \theta} > 0, \tag{SC.5}$$

$$\lim_{\lambda\to\kappa/\Pi}\overline{h} = \lim_{\lambda\to\infty}\overline{h} = 0. \tag{SC.6}$$

SC.2.2 General comparative statics

Consider the general case of $\gamma \geq 0$. Since $\sigma = 0$, Inequality (11) for $h = 0$ implies that $\underline{h} > 0$ if and only if

$$\rho\overline{h} < \frac{(1-\theta)\gamma}{\rho+\lambda+(1-\theta)\gamma}(\lambda\Pi - \kappa). \tag{SC.7}$$

By Section SC.1, the following result is obtained.

Proposition SC.2.2 (Startup financing)**.** *Denote*

$$\eta \equiv \frac{(1-\theta)\gamma}{\rho+\lambda+(1-\theta)\gamma}, \quad \xi \equiv \frac{\lambda\Pi - \kappa}{\rho}.$$

The equilibrium is characterized by $\underline{h} = 0$ *and* $\overline{h}$ *implicitly defined by Equation (SC.3) if*

$$\theta\rho\big((\rho+\lambda)\eta\xi+\kappa\big)+(1-\theta)\lambda\big((\rho+\lambda)\Pi-\kappa\big) \geq \rho\kappa\exp\left(\frac{\rho+\lambda}{\kappa}\eta\xi\right). \tag{SC.8}$$

If the inequality is strictly reversed, then $\underline{h} = \overline{h} - \Delta h > 0$ *and* $\overline{h} > \Delta h$ *is implicitly defined by*

$$\frac{1-\theta}{\rho+\lambda}\left((\rho+\lambda+\theta\gamma)\lambda\Pi - \theta\rho\gamma\left(\overline{h}+\frac{\kappa}{\rho+\lambda+\theta\gamma}\right)-(\lambda+\theta\gamma)\kappa\right)$$
$$= \rho\left(\frac{\rho+\lambda+\gamma}{\gamma}\Delta h + (1-\theta)\frac{\kappa}{\rho+\lambda+\theta\gamma}\right)\exp\left(\frac{\rho+\lambda+\theta\gamma}{\kappa}(\overline{h}-\Delta h)\right), \tag{SC.9}$$

where $\Delta h = \overline{h} - \underline{h} > 0$ *is given by*

$$1+\frac{\rho+\lambda}{\kappa}\left(1+\frac{\rho+\lambda}{\gamma}\right)\Delta h = \exp\left(\frac{\rho+\lambda}{\kappa}\Delta h\right). \tag{SC.10}$$

Proposition SC.2.3 (Comparative statics – startups)**.** $\overline{h}$ *strictly increases in* Π. *When* $\underline{h} > 0$, $\underline{h}$ *strictly increases in* Π *and* Δh *is constant in* Π *and strictly decreasing in* λ. *Last,* $\overline{h}$, $\underline{h}$ *and* Δh *converge to zero as either (i)* Π *goes to* κ/λ *or (ii)* λ *goes to either* κ/Π *or* ∞.

Proposition SC.2.4 (Breakeven re-inclusion – Startups)**.** $\underline{\gamma}$ *strictly decreases in* Π, *and converges to zero as* Π *goes to* ∞. *It goes to* ∞ *as* Π *goes down to* κ/λ, *the lower bound in Assumption SC.2.1.*

SC.3 Proofs

Proposition SC.2.1 (Comparative statics – Startups without re-inclusion).

Proof. Most results above are straightforward. For the first inequality in (SC.4), it is sufficient to show that $\lambda\Pi - \kappa - \rho\overline{h}(0) > 0$, where $\overline{h}(\theta)$ is $\overline{h}$ expressed as a function of $\theta \in [0, 1]$.

To prove the claim, note that

$$
\begin{aligned}
& \lambda\Pi - \kappa > \rho\overline{h}(0) \\
\iff \quad & \exp\left(\frac{\rho+\lambda}{\kappa}\frac{\lambda\Pi-\kappa}{\rho}\right) > \exp\left(\frac{\rho+\lambda}{\kappa}\overline{h}(0)\right) \\
& \qquad = \frac{\lambda}{\rho}\left((\rho+\lambda)\frac{\Pi}{\kappa} - 1\right) \qquad \because (SC.3) \text{ with } \theta \equiv 0 \\
& \qquad = 1 + \frac{\rho+\lambda}{\kappa}\frac{\lambda\Pi-\kappa}{\rho} \\
\iff \quad & \frac{\rho+\lambda}{\kappa}\frac{\lambda\Pi-\kappa}{\rho} > 0,
\end{aligned}
$$

which is equivalent to Assumption SC.2.1. □

Proposition SC.2.2 (Startup financing).

Proof. If (SC.7) holds, then Inequality (SC.7) also holds with $\overline{h}$ defined by (SC.3). Therefore, $\underline{h} = 0$. Now, consider the case where (SC.7) fails. Let us use Smooth Pasting and Stationary Recursion to determine $(\underline{h}, \overline{h})$.

Smooth Pasting. For ease of notation, denote $\underline{V}_o \equiv V_o(\underline{h})$ and $\underline{V} \equiv \theta(V(\overline{h}) - \overline{h} + \underline{h}) + (1-\theta)\underline{V}_o$. The insiders' non-excluded and excluded value functions for $h \in [\underline{h}, \overline{h}]$ are given by

$$
\begin{aligned}
V(h) &= \int_0^{(h-\underline{h})/\kappa} \lambda e^{-(\rho+\lambda)t}\big(\Pi + h - \kappa t\big)\, dt + e^{-\frac{\rho+\lambda}{\kappa}(h-\underline{h})}\underline{V} \\
&= \frac{\lambda}{\rho+\lambda}\left(\Pi + h - \frac{\kappa}{\rho+\lambda}\right) - \left(\frac{\lambda}{\rho+\lambda}\left(\Pi + \underline{h} - \frac{\kappa}{\rho+\lambda}\right) - \underline{V}\right) e^{-\frac{\rho+\lambda}{\kappa}(h-\underline{h})}, \\
V_o(h) &= V(h) - e^{-\frac{\rho+\lambda+\gamma}{\kappa}(h-\underline{h})}\big(\underline{V} - \underline{V}_o\big).
\end{aligned}
$$

$V_o(h)$ is derived based on the observation that, given the strategy of waiting on $(\underline{h}, h)$ regardless of market access, the only difference that exclusion creates is that one has $\underline{V}_o$ instead of $\underline{V}$ at $h = \underline{h}$ if neither success nor re-inclusion occurs while the internal funds h run down to $\underline{h}$.

Note that

$$
\begin{aligned}
& V(\overline{h}) = \frac{1}{\rho+\lambda}\big(\lambda(\Pi + \overline{h}) - \kappa\big) \\
\implies & \frac{\rho\kappa}{(\rho+\lambda)^2} = \left(\frac{\lambda}{\rho+\lambda}\left(\Pi + \underline{h} - \frac{\kappa}{\rho+\lambda}\right) - \underline{V}\right)\exp\left(-\frac{\rho+\lambda}{\kappa}(\overline{h} - \underline{h})\right) \\
\implies & V(h) = \frac{\lambda}{\rho+\lambda}\left(\Pi + h - \frac{\kappa}{\rho+\lambda}\right) - \frac{\rho\kappa}{(\rho+\lambda)^2}\exp\left(\frac{\rho+\lambda}{\kappa}(\overline{h} - h)\right), \text{ and} \\
& \underline{V} - \underline{V}_o = \frac{\theta}{1-\theta}\left(-\frac{\rho}{\rho+\lambda}\left(\overline{h} - \underline{h} + \frac{1}{\rho+\lambda}\kappa\right) + \frac{\rho\kappa}{(\rho+\lambda)^2}\exp\left(\frac{\rho+\lambda}{\kappa}(\overline{h} - \underline{h})\right)\right).
\end{aligned}
$$

Next, denote by V_d a payoff function on $(\underline{h}, \overline{h}]$ for the deviation strategy of immediate financing. That is, for $h \in (\underline{h}, \overline{h}]$,

$$V_d(h) \equiv \theta\big(V(\overline{h}) - \overline{h} + h\big) + (1-\theta)V_o(h).$$

Smooth pasting condition is

$$V'(\underline{h}) = V_d'(\underline{h}) = \theta + (1-\theta)\left(V'(\underline{h}) + \frac{\rho+\lambda+\gamma}{\kappa}(\underline{V} - \underline{V}_o)\right),$$

which, after some algebra, is equivalent to Equation (SC.10).

Stationary Recursion. This time, start by deriving V_o on $[0, \underline{h}]$, which satisfies:

$$\begin{aligned}
&\rho V_o(h) = \gamma\Big(\theta\big(V(\overline{h}) - \overline{h} + h\big) + (1-\theta)V_o(h) - V_o(h)\Big) + \lambda\big(\Pi + h - V_o(h)\big) - \kappa V_o'(h), \\
&V_o(0) = 0 \\
\implies\ & V_o(h) = \frac{1}{\rho+\lambda+\theta\gamma}\Bigg[\lambda\Pi + \theta\gamma\big(V(\overline{h}) - \overline{h}\big) - \frac{\lambda+\theta\gamma}{\rho+\lambda+\theta\gamma}\kappa + (\lambda+\theta\gamma)h \\
&\qquad - \left(\lambda\Pi + \theta\gamma\big(V(\overline{h}) - \overline{h}\big) - \frac{\lambda+\theta\gamma}{\rho+\lambda+\theta\gamma}\kappa\right)\exp\left(-\frac{\rho+\lambda+\theta\gamma}{\kappa}h\right)\Bigg].
\end{aligned}$$

Since $\Delta h \equiv \overline{h} - \underline{h} > 0$ has been determined by Equation (SC.10), $\overline{h}$ is obtained by the recursion:

$$V(\overline{h} - \Delta h) = \theta\Big(V(\overline{h}) - \Delta h\Big) + (1-\theta)V_o(\overline{h} - \Delta h).$$

Simplifying and substituting (SC.10) give Equation (SC.9). □

Proposition SC.2.3 (Comparative statics – startups)

Proof. Immediate from Proposition SC.2.2 via implicit differentiation. □

Proposition SC.2.4 (Breakeven re-inclusion – Startups).

Proof. $\underline{\gamma}$ is defined by $\underline{\eta}\xi = \overline{h}^*$, where $(\underline{h}^*, \overline{h}^*)$ is the equilibrium associated with $\gamma = \underline{\gamma}$ and $(\underline{\eta}, \xi)$ given by Proposition SC.2.2. Since $\gamma = \underline{\gamma}$ implies $\underline{h}^* = 0$, Proposition SC.2.1, in particular Equation (SC.5), holds with $\overline{h}$ replaced with $\underline{\eta}\xi$. Note that

$$\begin{aligned}
&\frac{\lambda}{\rho}\frac{1-\theta}{\exp\left(\frac{\rho+\lambda}{\kappa}\underline{\eta}\xi\right) - \theta} = \frac{\partial(\underline{\eta}\xi)}{\partial\Pi} = \underline{\eta}\frac{\partial\xi}{\partial\Pi} + \xi\frac{\partial\underline{\eta}}{\partial\Pi}, \quad \frac{\partial\xi}{\partial\Pi} = \frac{\lambda}{\rho} \\
\implies\ & \frac{\partial\underline{\eta}}{\partial\Pi} = \frac{\lambda}{\lambda\Pi - \kappa}\left(\frac{1-\theta}{\exp\left(\frac{\rho+\lambda}{\kappa}\underline{\eta}\xi\right) - \theta} - \frac{1-\theta}{1 + \frac{\rho+\lambda}{\underline{\gamma}} - \theta}\right).
\end{aligned}$$

Since $\overline{h}^* = \underline{\eta}\xi$, smooth pasting holds at $\underline{h}^* = 0$. Therefore, from Equation (SC.10),

$$\exp\left(\frac{\rho+\lambda}{\kappa}\underline{\eta}\xi\right) = 1 + \frac{\rho+\lambda}{\gamma}\frac{\rho+\lambda+\gamma}{\kappa}\underline{\eta}\xi.$$

Next, assume for now that $\underline{\eta}\xi = \overline{h}^* > \frac{\kappa}{\rho+\lambda+\underline{\gamma}}$, which will be established at the end. Then,

$$\exp\left(\frac{\rho+\lambda}{\kappa}\underline{\eta}\xi\right) > 1 + \frac{\rho+\lambda}{\underline{\gamma}},$$

Thus, $\partial\underline{\eta}/\partial\Pi < 0$. Since $\underline{\eta} = \frac{(1-\theta)\underline{\gamma}}{\rho+\lambda+(1-\theta)\underline{\gamma}}$, we have $\partial\underline{\gamma}/\partial\Pi < 0$.

Next is the convergence claim. Since $\overline{h}^*$ satisfies Equation (SC.3), it goes to ∞ as Π does. Note that $\Delta h^* \equiv \overline{h}^* - \underline{h}^* = \overline{h}^*$ satisfies Equation (SC.10) with $\gamma = \underline{\gamma}$. Since $\rho,\ \lambda,\ \kappa$ are fixed, the only way for the solution of Equation (SC.10) to be satisfied by a Δh that diverges to infinity is by having the linear coefficient on the LHS also diverge to infinity. This can only be achieved if $\underline{\gamma}$ goes to zero, as claimed.

The divergence claim is straightforward from $\underline{\eta}\xi > \frac{\kappa}{\rho+\lambda+\underline{\gamma}}$. Since $\xi = (\lambda\Pi - \kappa)/\rho$ and $\underline{\eta}$ is in the unit interval, the LHS vanishes as Π goes down to κ/λ. Therefore, the RHS also vanishes, i.e., $\underline{\gamma} \to \infty$.

Finally, as for the intermediate claim on the strict lower bound on $\underline{\eta}\xi$, first rearrange the Smooth Pasting condition – i.e., Equation (SC.10) – into the following:

$$\frac{\gamma}{\rho+\lambda}\left(\exp\left(\frac{\rho+\lambda}{\kappa}\Delta h\right) - 1\right) = \frac{\rho+\lambda+\gamma}{\kappa}\Delta h.$$

Denote the LHS and RHS above as functions of Δh. Note that LHS(0) = RHS(0) and LHS$'(0) <$ RHS$'(0)$. Therefore, the LHS crosses the RHS only once and from below on $\mathbb{R}_{++}$. Note that

$$\text{LHS}\left(\frac{\kappa}{\rho+\lambda+\gamma}\right) = \frac{\gamma}{\rho+\lambda}\left(\exp\left(\frac{\rho+\lambda}{\rho+\lambda+\gamma}\right) - 1\right) < 1 = \text{RHS}\left(\frac{\kappa}{\rho+\lambda+\gamma}\right).$$

This holds for any set of parameters because, letting $f(x) \equiv x\left(\exp\left(\frac{1}{1+x}\right) - 1\right)$, we have

$$\forall\ x > 0,\ f'(x) > 0, \text{ and } \lim_{x\to\infty} f(x) = 1.$$

Therefore, LHS$(\Delta h) <$ RHS(Δh) for any $\Delta h \in (0, \frac{\kappa}{\rho+\lambda+\gamma}]$. That is, if Smooth Pasting holds at $\underline{h}$, then it must be that $\Delta h = \overline{h} - \underline{h} > \frac{\kappa}{\rho+\lambda+\gamma}$. Since $\gamma = \underline{\gamma}$ means that Smooth Pasting holds at $\underline{h}^* = 0$, it must be that $\underline{\gamma}\xi = \overline{h}^* = \Delta h^* > \frac{\kappa}{\rho+\lambda+\underline{\gamma}}$, as claimed. □